



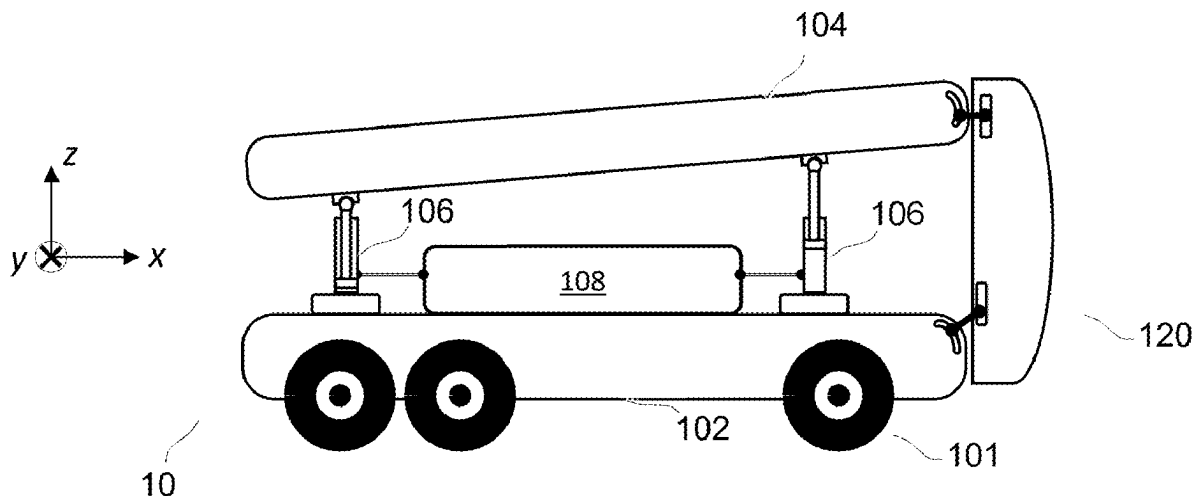
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(19) **United States**(12) **Patent Application Publication** (10) **Pub. No.: US 2025/0282492 A1**
(43) **Pub. Date: Sep. 11, 2025**(54) **GROUND VEHICLE SYSTEM FOR
HORIZONTALLY LANDING, LAUNCHING
AND TAXING OF AIRCRAFT AND
SPACECRAFT**(52) **U.S. Cl.**
CPC *B64F 1/227* (2013.01); *B64F 1/0299*
(2020.01); *B64F 1/228* (2013.01)(71) Applicant: **DESTINUS SA**, Payerne (CH)(57) **ABSTRACT**(72) Inventors: **Sergiu Petre ILIEV**, Payerne (CH);
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Mikhail KOKORICH, Payerne (CH)(21) Appl. No.: **18/861,484**(22) PCT Filed: **Apr. 29, 2022**(86) PCT No.: **PCT/EP2022/061537**

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The invention is directed to an aircraft carrier ground vehicle, which comprises a chassis, a landing platform, a coupling mechanism, and an actuation mechanism. The landing platform is designed to accommodate an aircraft. The coupling mechanism connects the landing platform to the chassis to form a kinematic pair, in which the landing platform is rotatably mounted on the chassis. The actuation mechanism is adapted to rotate the landing platform relative to the chassis, to allow an aircraft to land on the landing platform as the vehicle moves along a landing direction of the aircraft and compensate for an inclination angle of the aircraft on landing, in operation. The aircraft carrier ground vehicles further allow taxiing and take-off of the aircraft. The invention is further directed to related aircraft carrier ground vehicle systems, which include several vehicles that are swarmed for landing an aircraft, and aircraft landing and take-off methods.



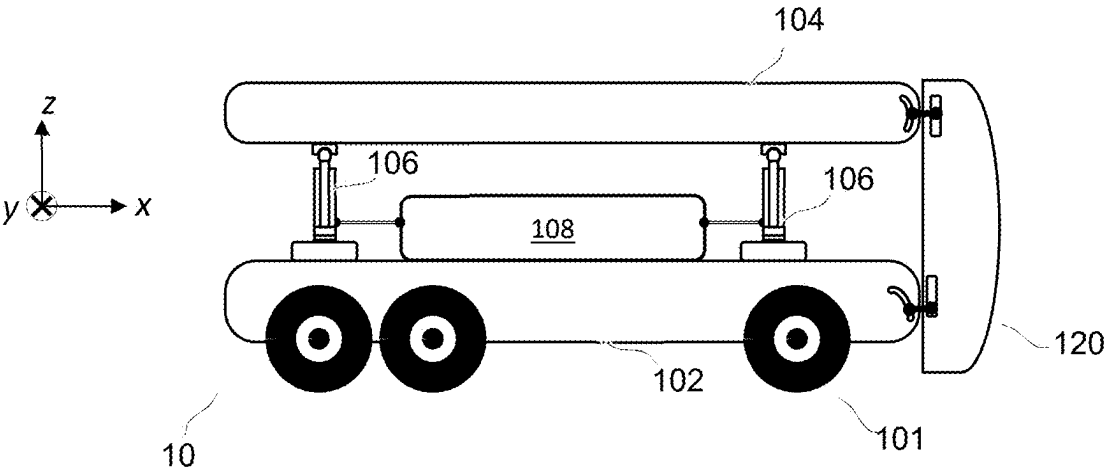


FIG. 1

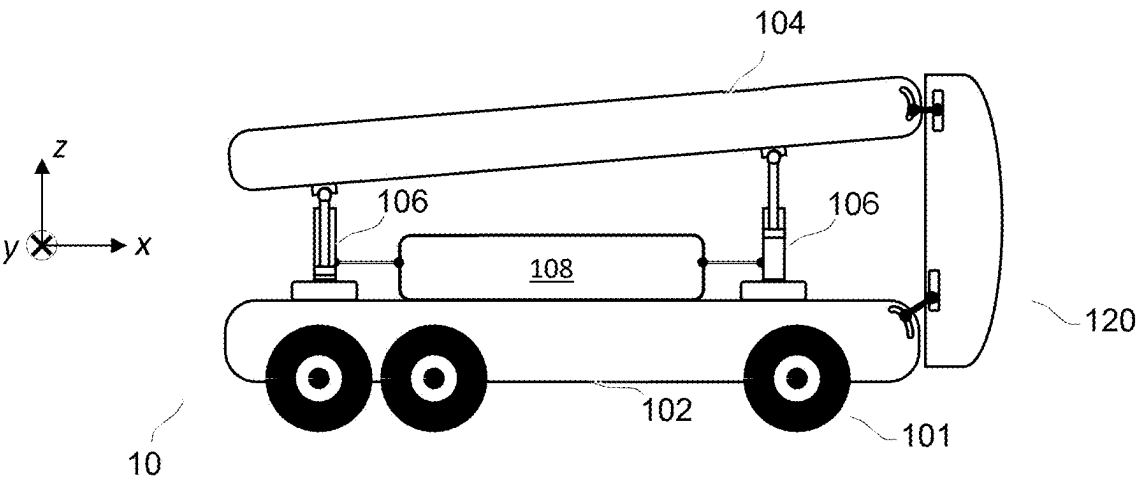


FIG. 2

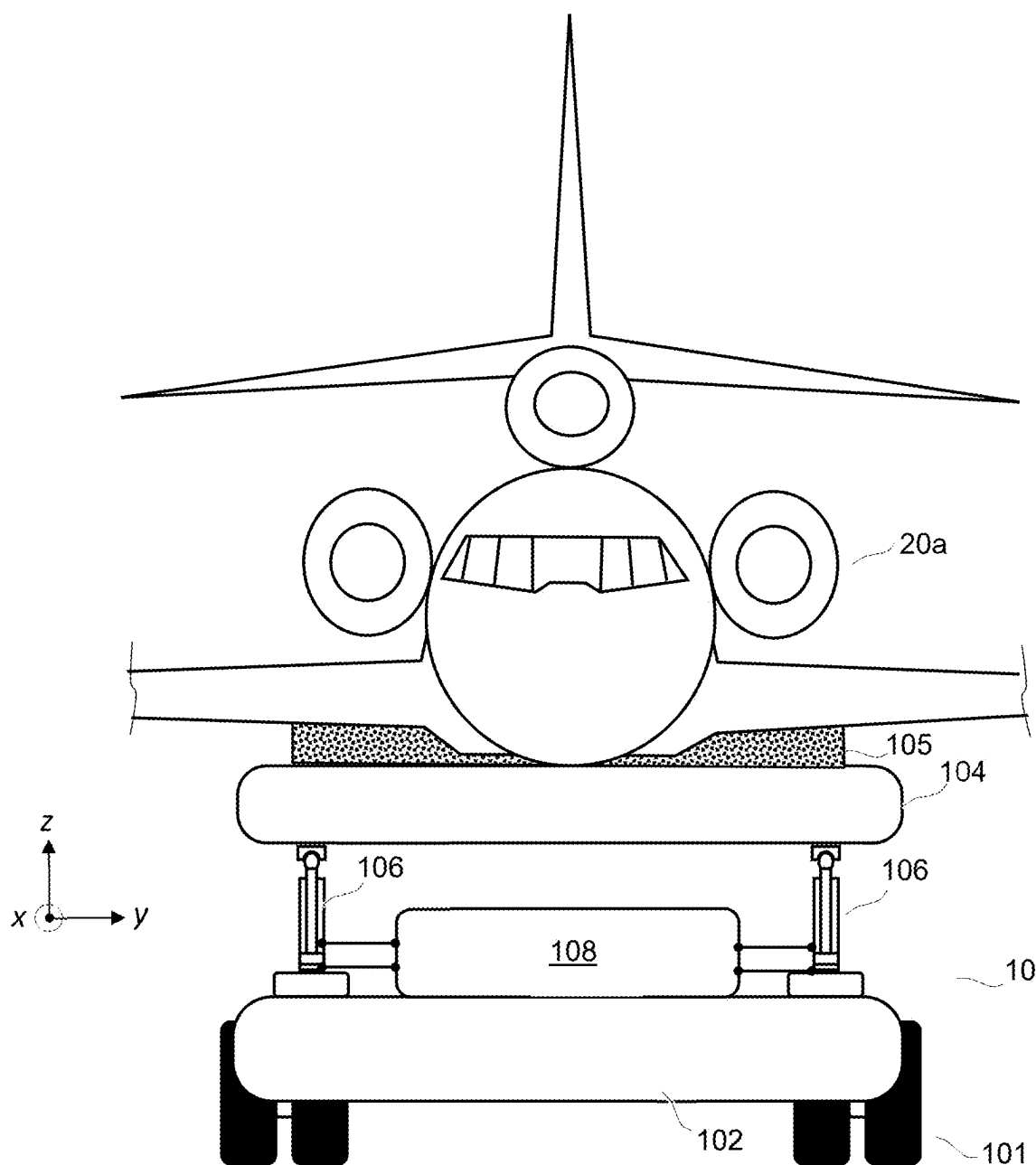


FIG. 3

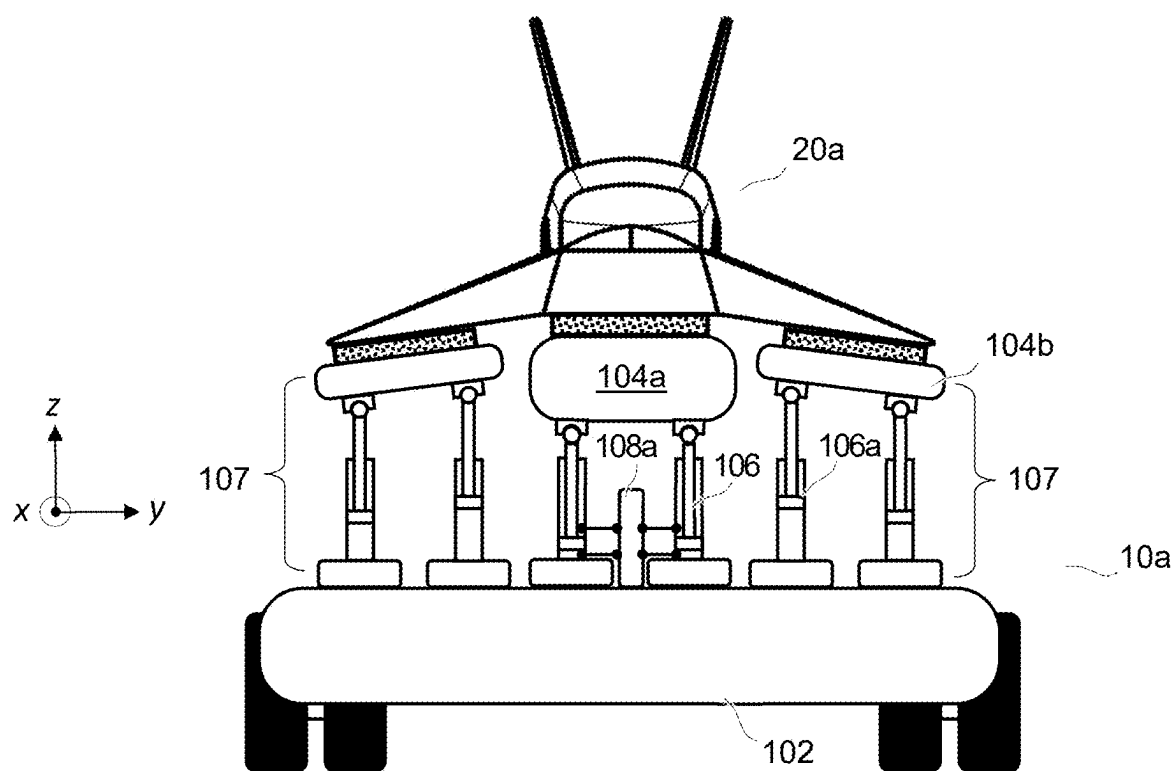


FIG. 4A

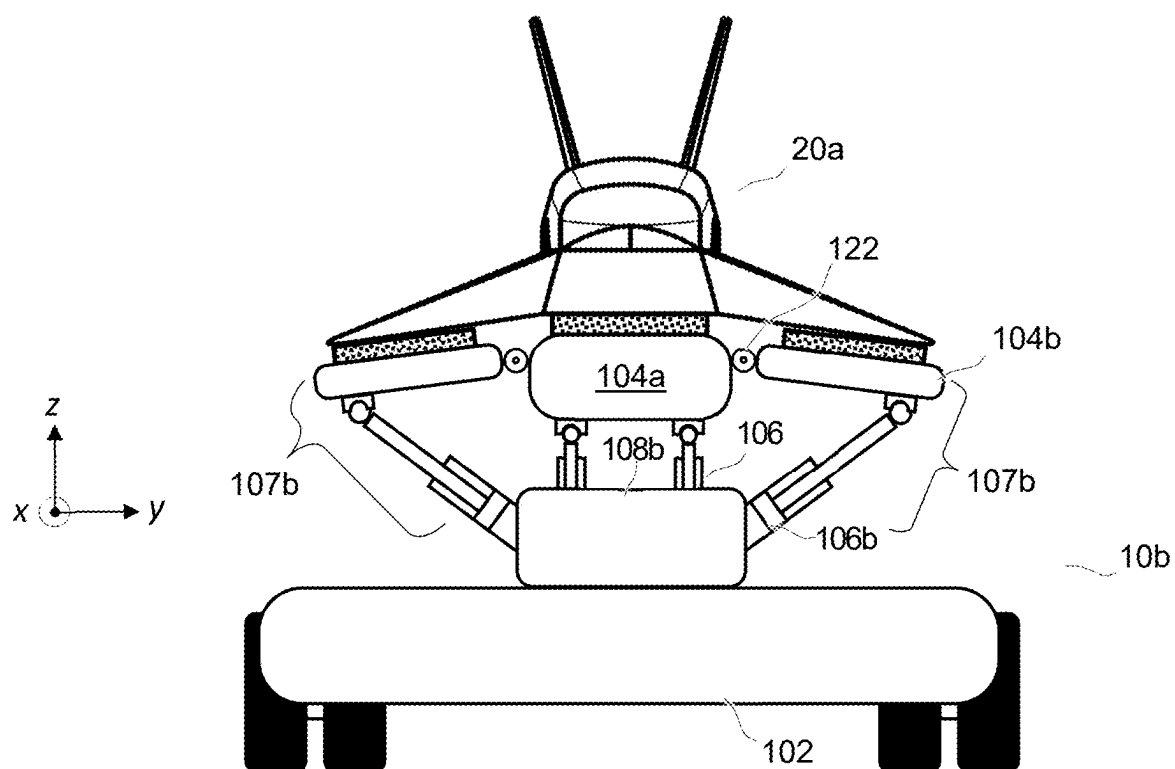


FIG. 4B

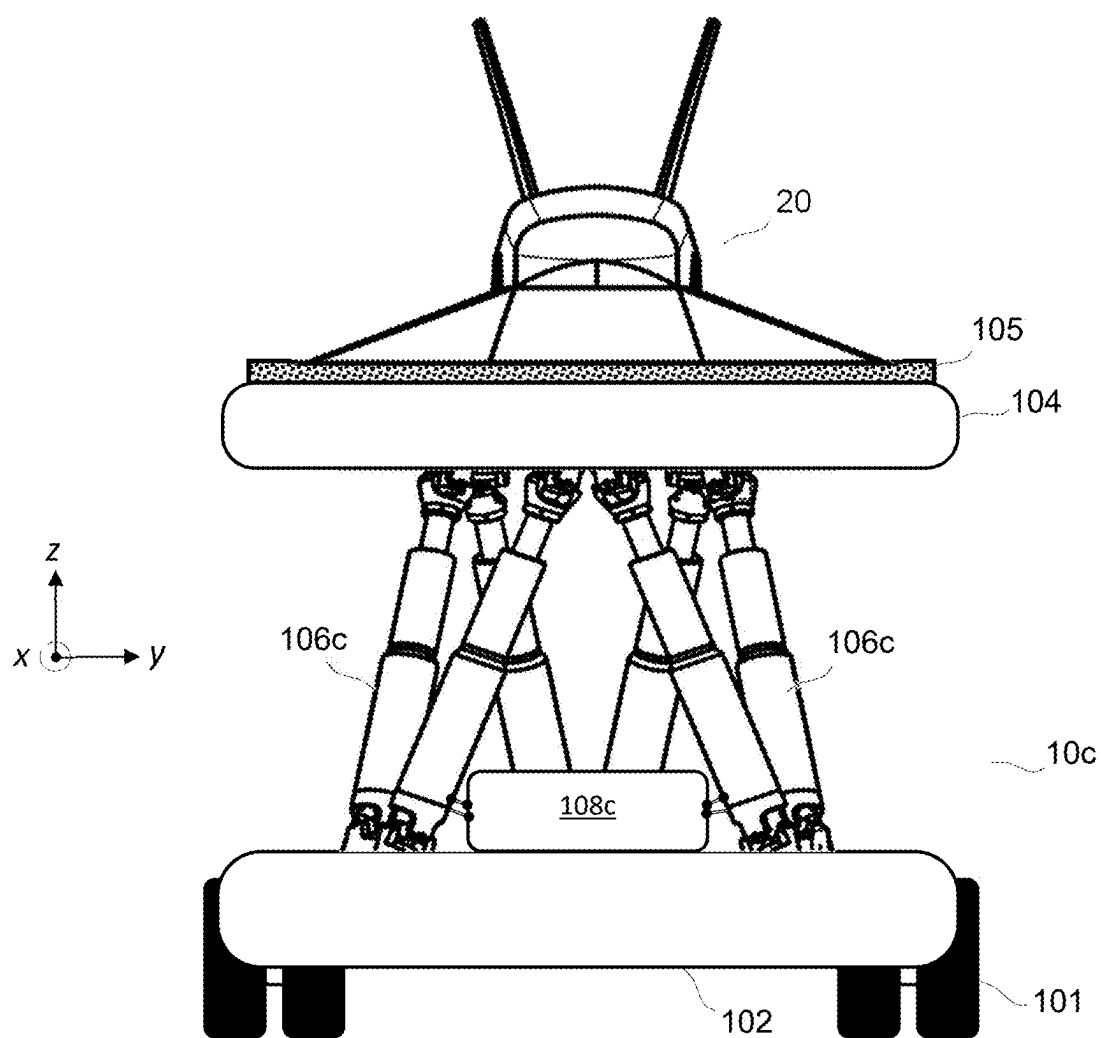


FIG. 5

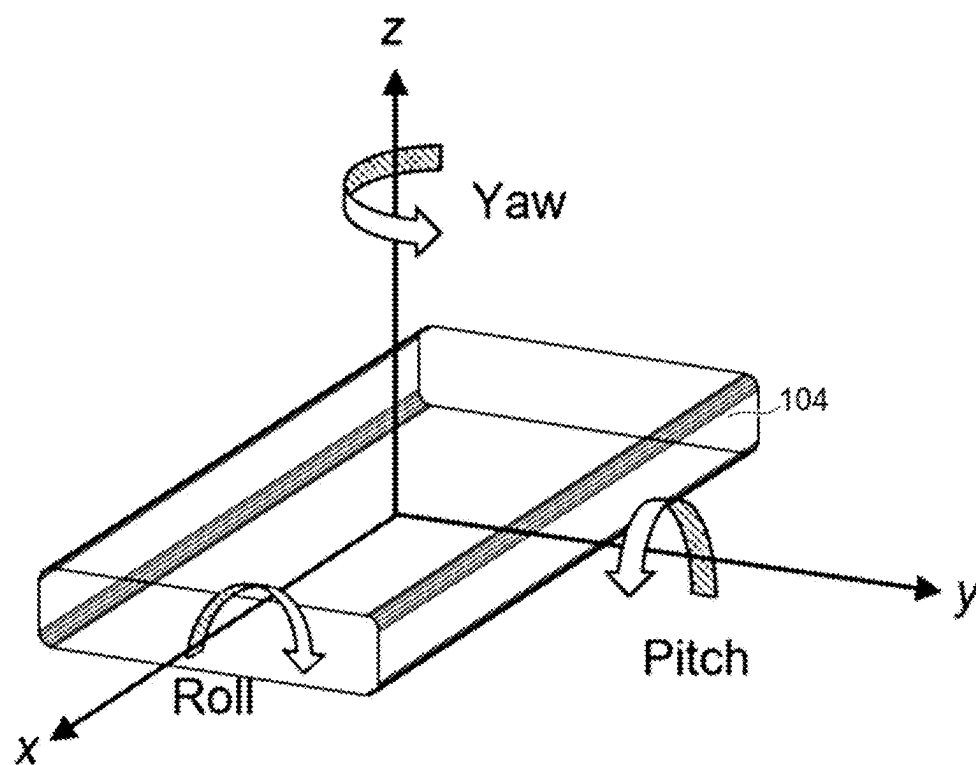


FIG. 6

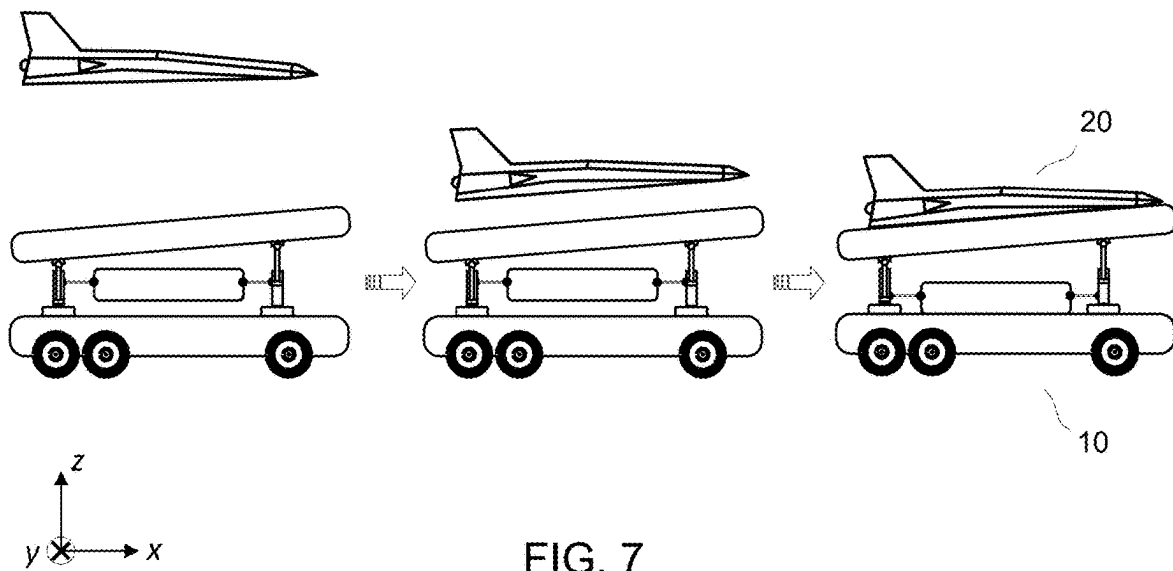


FIG. 7

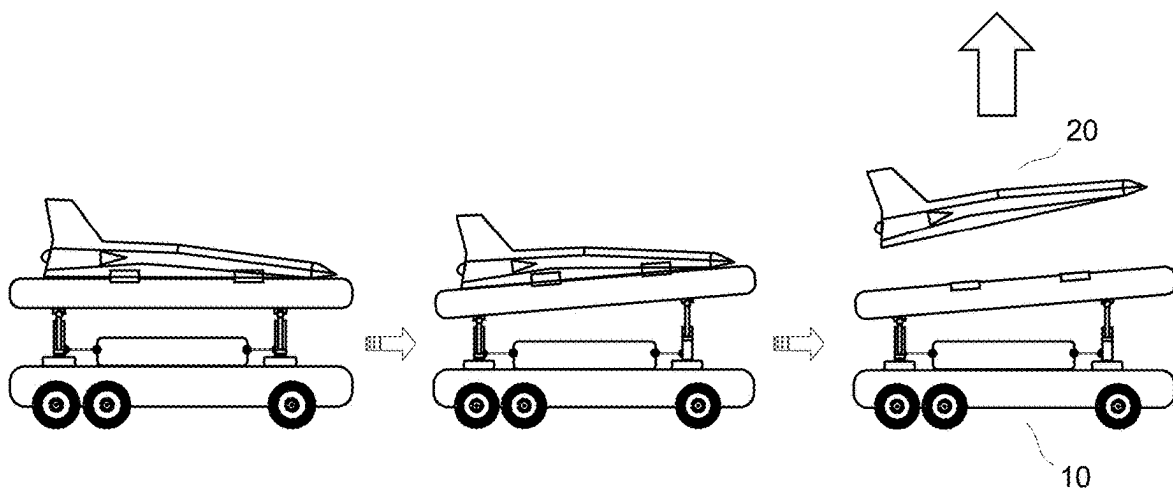


FIG. 8

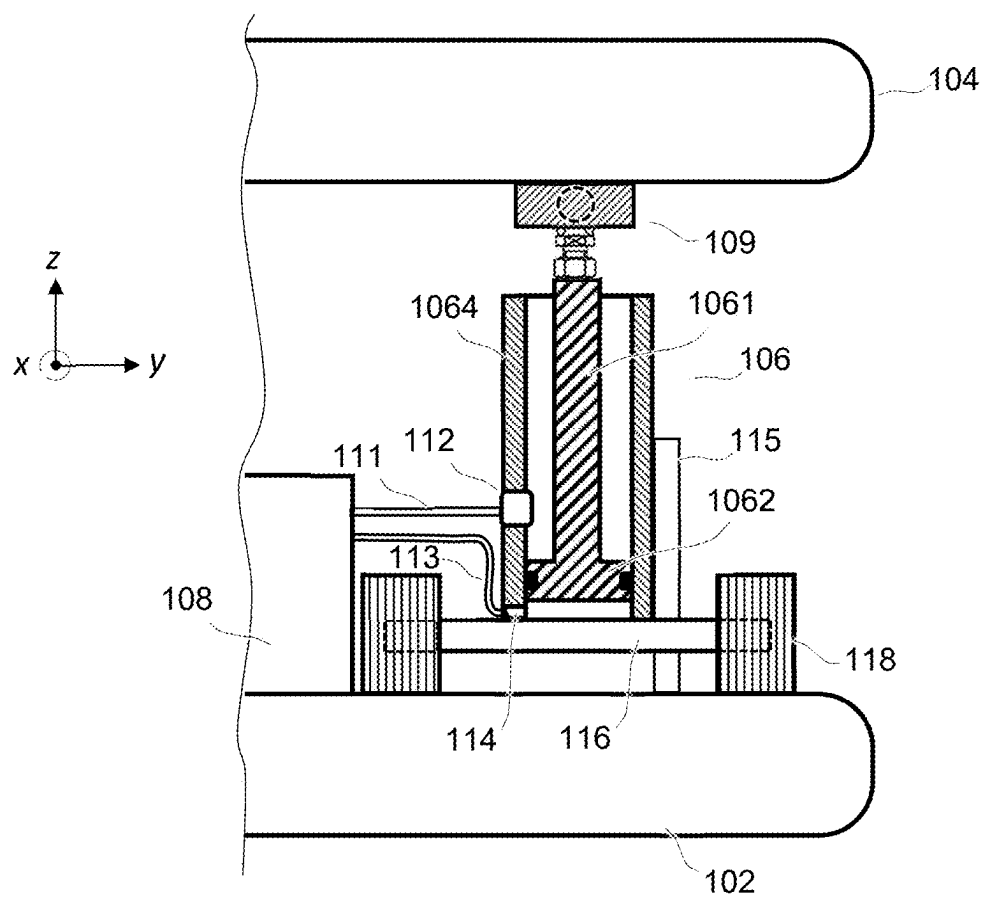


FIG. 9

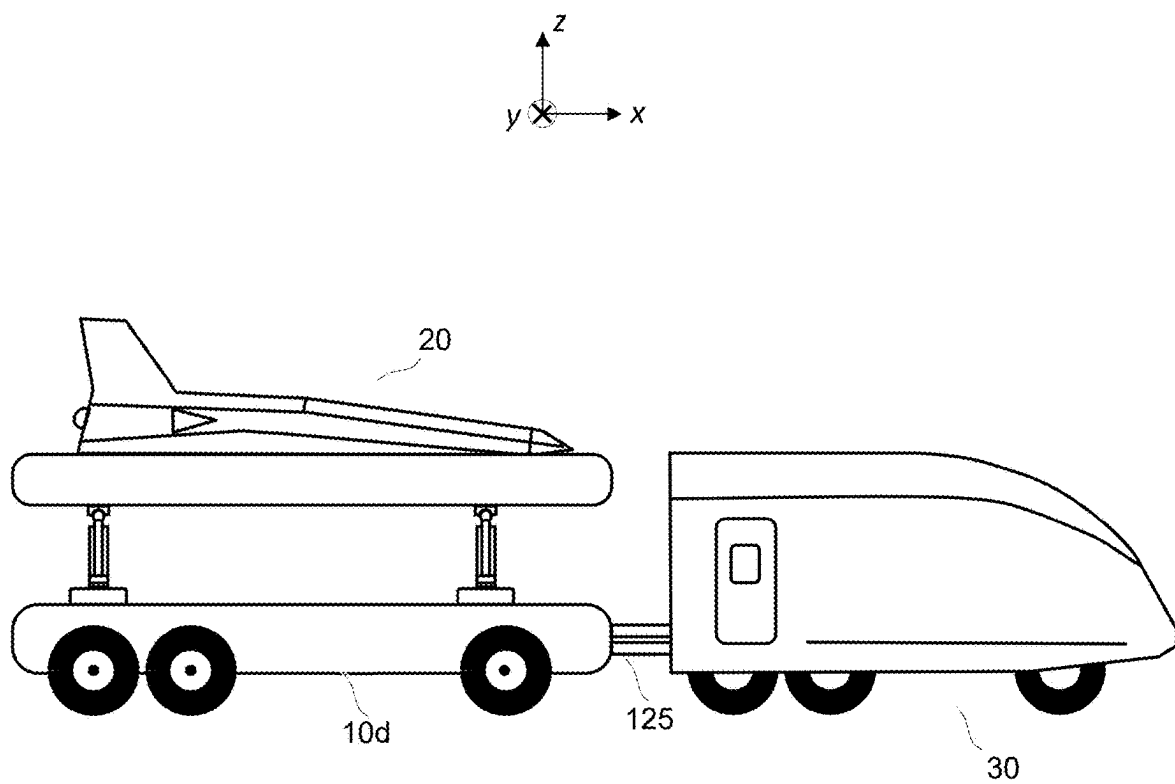


FIG. 10

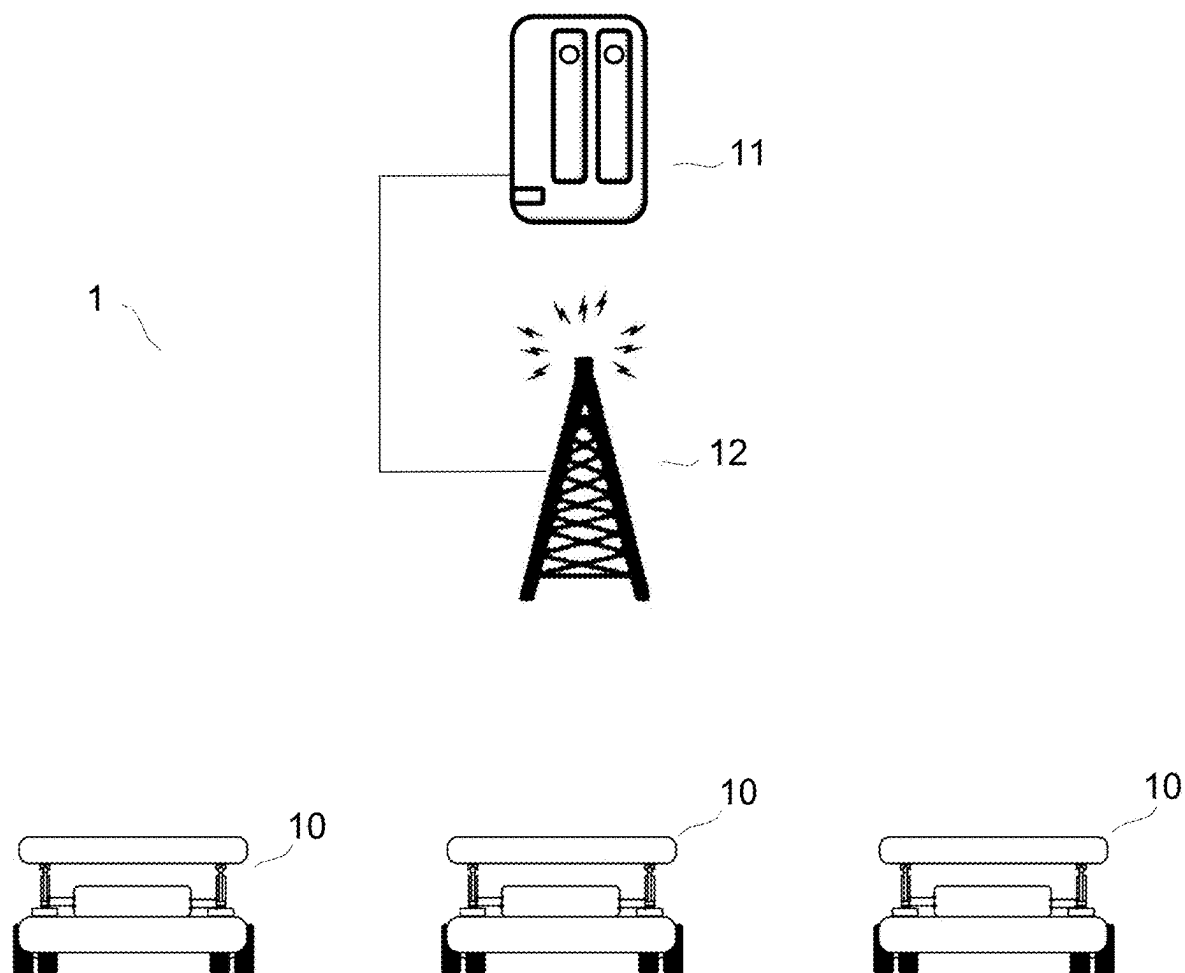


FIG. 11

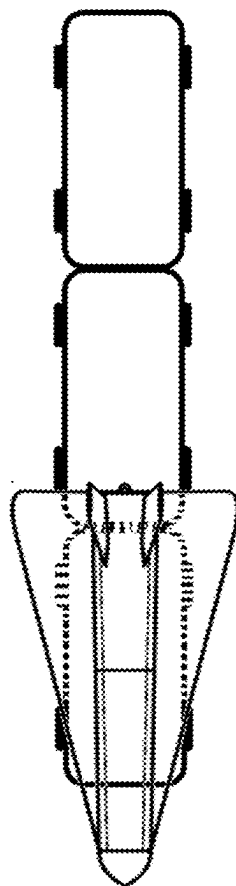
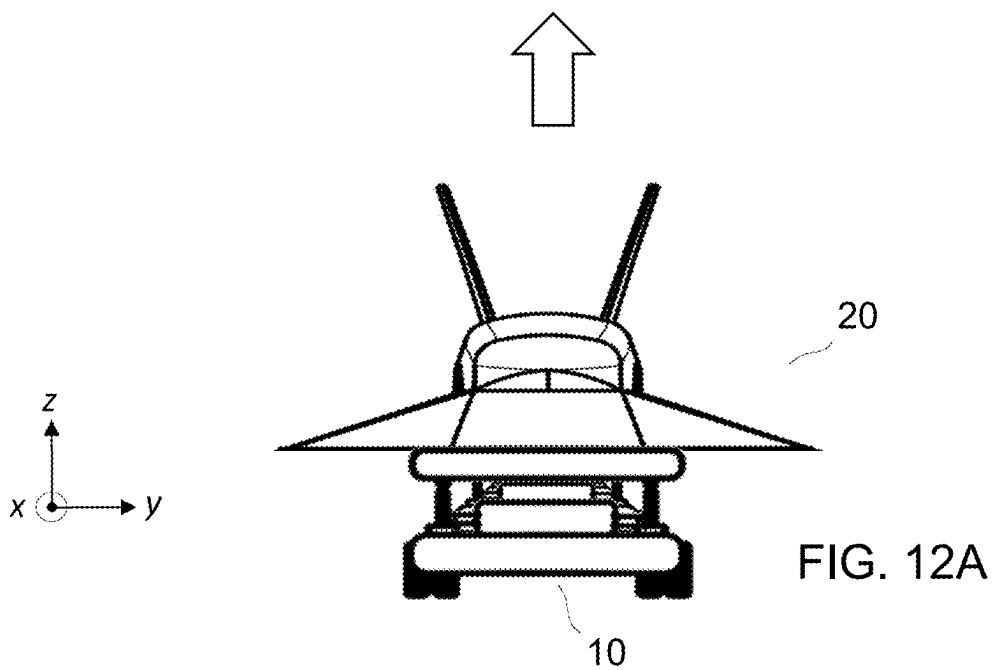


FIG. 12B

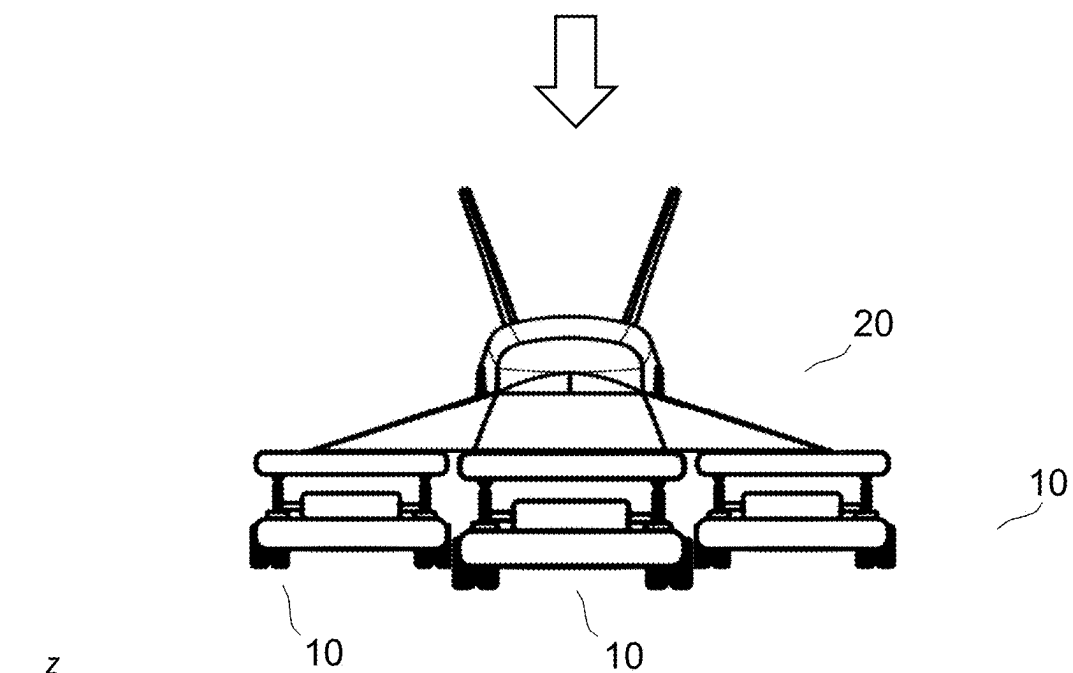


FIG. 12C

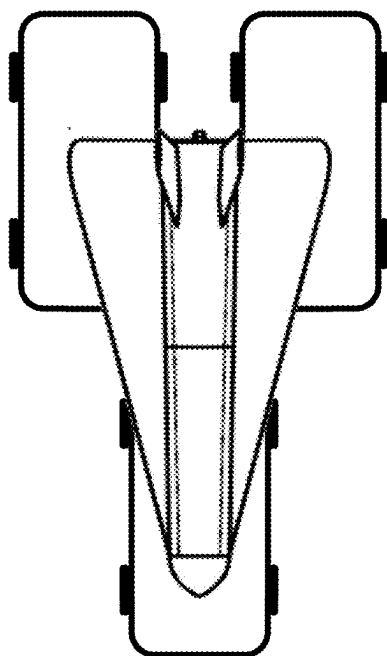


FIG. 12D

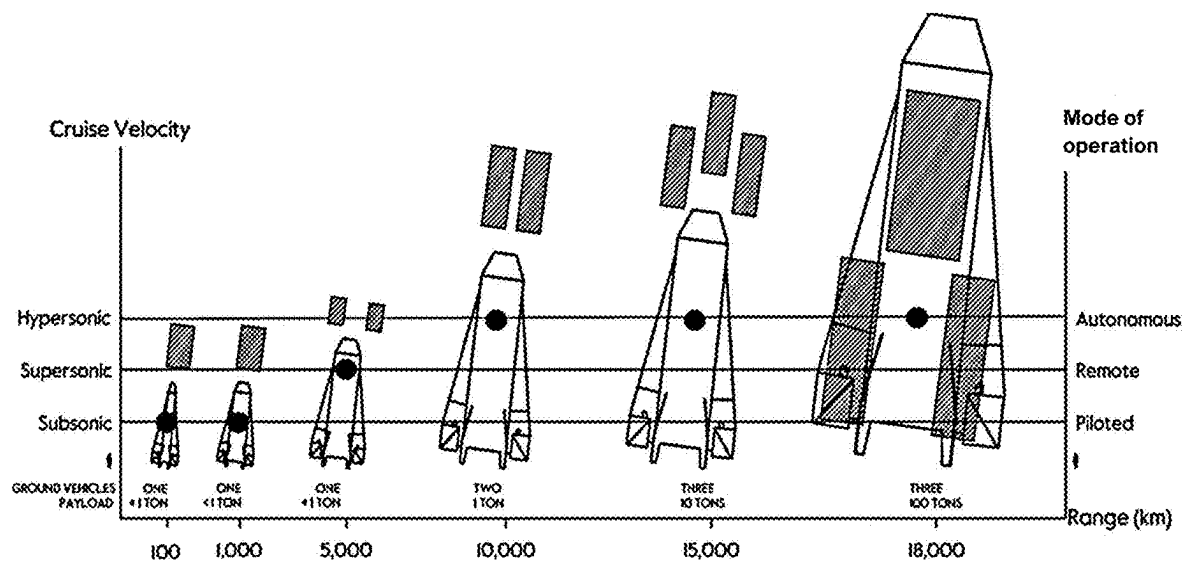


FIG. 13

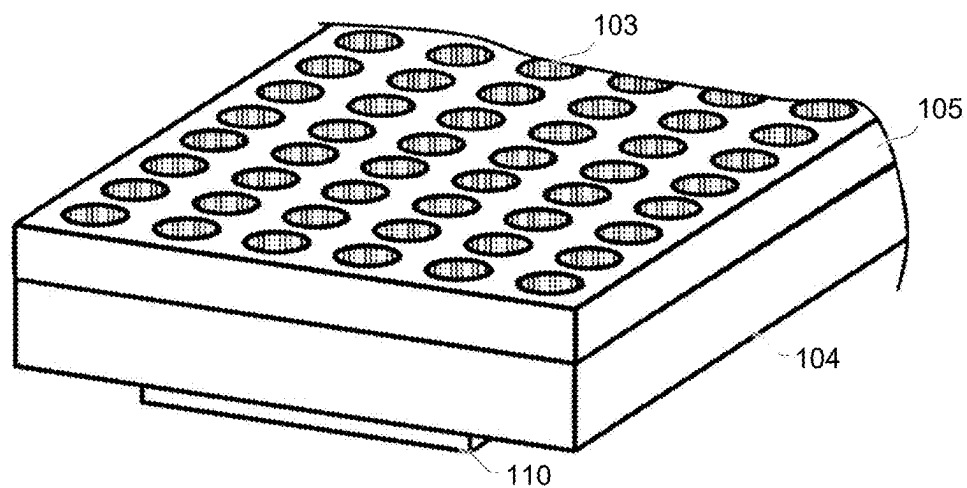


FIG. 14

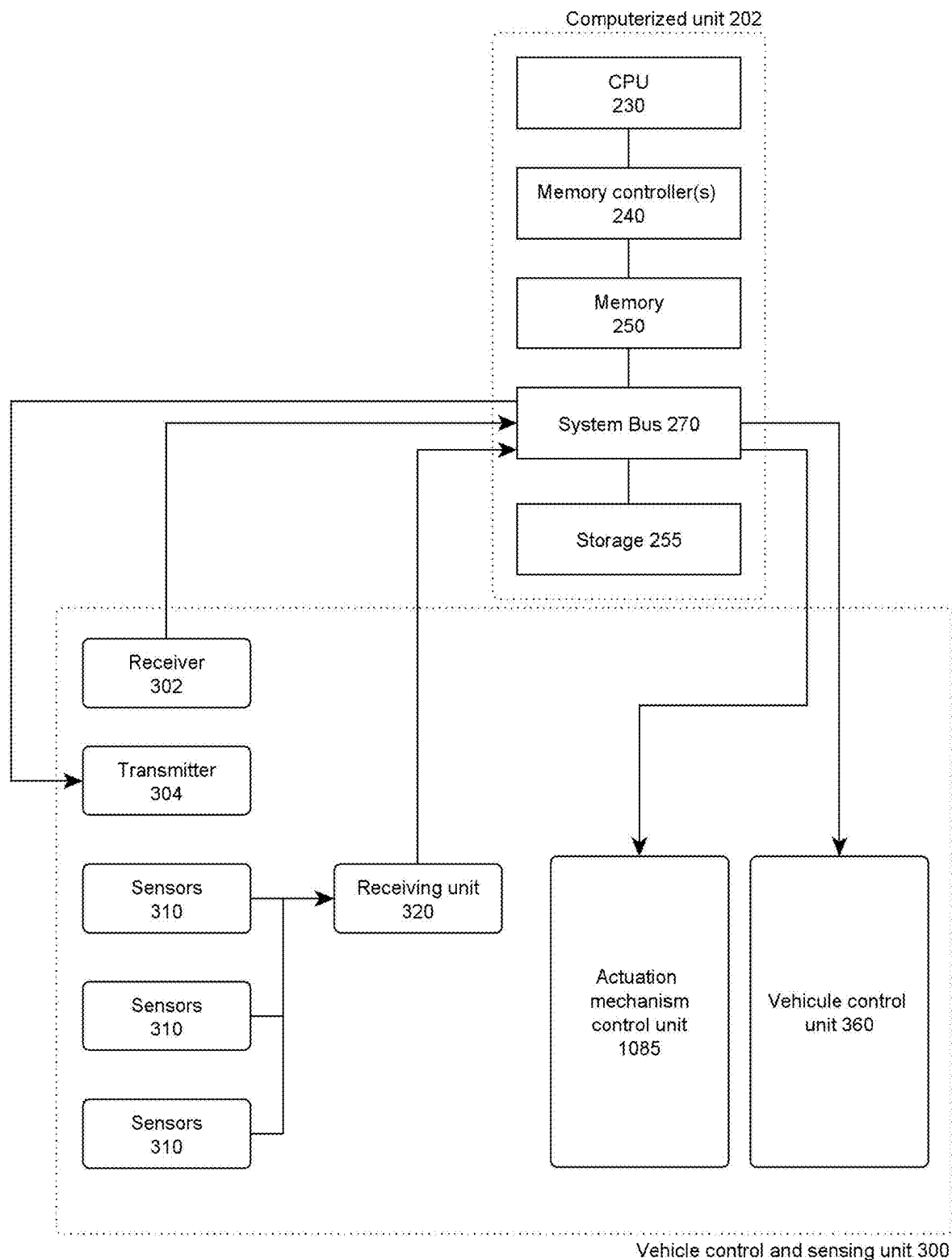


FIG. 15

**GROUND VEHICLE SYSTEM FOR
HORIZONTALLY LANDING, LAUNCHING
AND TAXING OF AIRCRAFT AND
SPACECRAFT**

BACKGROUND

[0001] The invention relates in general to aircraft carrier ground vehicles, as well as to aircraft landing and take-off methods based on such vehicles. In particular, the invention concerns a ground vehicle comprising a landing platform rotatably mounted on a chassis.

[0002] Aircraft landing gears are only utilized during take-off, landing, and taxiing. They perform no useful function in flight. Therefore, the landing gear results in unused mass and volume for much of the aircraft mission. Taking into account the structural reinforcement required by the landing gear, the additional mass induced by a landing gear may typically represent between 5 to 10% of the total mass of the aircraft.

[0003] The purpose of the invention is to minimise the physical constraints resulting from the landing gear in the design and fabrication of an aircraft and thus improve the performance of the aircraft.

SUMMARY

[0004] According to a first aspect, the present invention is embodied as an aircraft carrier ground vehicle, which comprises a chassis, a landing platform, a coupling mechanism, and an actuation mechanism. The landing platform is designed to accommodate an aircraft. The coupling mechanism connects the landing platform to the chassis to form a kinematic pair, in which the landing platform is rotatably mounted on the chassis. The actuation mechanism is adapted to rotate the landing platform relative to the chassis (i.e., in accordance with the coupling mechanism), to allow an aircraft to land on the landing platform as the vehicle moves along a landing direction of the aircraft and compensate for an inclination angle of the aircraft on landing, in operation. The aircraft carrier ground vehicles may further be designed to allow taxiing and take-off of the aircraft.

[0005] The invention proposes an aircraft carrier ground vehicle equipped with a pivotable landing platform, which can be regarded as a remote landing gear. Thus, the aircraft does not need to be permanently equipped with a landing gear. This makes it possible to reduce the aircraft mass or, conversely, increase the payload of and volume available in the aircraft. Furthermore, this allows the structural efficiency of the aircraft to be optimized. In addition, the proposed solution allows an easier maintenance of the aircraft. Remarkably, rotating the platform makes it possible to compensate for the inclination angle of the aircraft and thus better absorb the shock on landing. For completeness, the present approach may be used to achieve smaller landing and take-off distances, e.g., by leveraging propulsion and breaking capabilities of the vehicle (or a tractor unit thereof), in addition to those of the aircraft.

[0006] In embodiments, the coupling mechanism is designed to allow an average plane of the landing platform to pivot by at most a first angle around an axis that is parallel to an axle direction of the vehicle. This makes it possible to accommodate for the aircraft pitch angle on landing.

[0007] Preferably, the coupling mechanism is further designed to allow the average plane of the landing platform

to pivot by at most a second angle around a longitudinal direction of the vehicle, which makes it possible to accommodate for the roll angle of the aircraft on landing. In preferred embodiments, the coupling mechanism includes a pneumatic suspension system having at least two pneumatic cylinders connecting the chassis to the landing platform. The pneumatic cylinders are actuable by the actuation mechanism to rotate the landing platform relative to the chassis, in operation. The actuation mechanism may for instance include pressurization means connected to each of the pneumatic cylinders, so as to independently operate the pneumatic cylinders as double-acting cylinders. Preferably, the pneumatic suspension system includes at least four pneumatic cylinders connecting the chassis to the landing platform. More preferably, the pneumatic suspension system includes six pneumatic cylinders; the platform and the pneumatic suspension system may for instance be jointly configured as a Stewart platform. Using pneumatic cylinders was found to provide satisfactory performance in terms of control, stability, and damping.

[0008] In embodiments, the coupling mechanism further comprises ball-and-socket joints linking piston rods of respective ones of the pneumatic cylinders to the landing platform, to prevent self-locking. In addition, the coupling mechanism further comprises revolute joints linking respective ones of the pneumatic cylinders to the chassis.

[0009] In preferred embodiments, a cylinder tube of each of the pneumatic cylinders is in thermal communication with a heat sink, the latter preferably forming part of the chassis. In variants, the heat sink is in thermal contact with the chassis. The aim is to rapidly absorb heat produced by rapidly compressing the pneumatic cylinders on touch down.

[0010] In embodiments, the landing platform further includes a deformable material layer, preferably a foam material layer. The deformable material layer extends on an exposed surface of the landing platform. This deformable layer provides an additional damping means, helps stabilizing the aircraft, and ensures friction between the aircraft and landing platform, hence working towards a coupling of the two surfaces.

[0011] Preferably, the vehicle further includes a pressurization system, and the platform further includes an array of suction cups that are at least partly embedded in the deformable material layer. The suction cups are connected to the pressurization system and are open on an exposed surface of the deformable material layer. The pressurization system is used to create a depression in the cups, allowing the cups to grip the aircraft on landing or maintain it before take-off. Conversely, the pressurization system may also be actuated to provoke a positive pressure surge, to help the aircraft to separate from the platform on take-off.

[0012] In embodiments, the vehicle further includes one or more auxiliary dampers, to best accommodate for the underside of the aircraft. Each auxiliary damper includes a landing pad, so as to allow the aircraft to land on both the landing platform and the one or more auxiliary dampers as the vehicle moves along the landing direction of the aircraft, in operation.

[0013] In embodiments, the vehicle further includes latching means adapted for gripping a structural part of the aircraft on landing. Preferably, the latching means includes a hard robotic mechanical latching, a soft robotic suction cup latching, and/or a magnetic attraction mechanism.

[0014] The vehicle may advantageously include a vehicle control system that comprises one or more control units coupled to the actuation mechanism to control a rotation of the landing platform relative to the chassis. The vehicle control system may further comprise one or more receivers coupled to the one or more control units, hence allowing remote control of the landing platform. The receivers are configured to receive instruction signals (e.g., from an external control unit) for the vehicle control system to forward corresponding control signals to the one or more control units, and accordingly control a rotation of the landing platform.

[0015] In embodiments, the vehicle is a powered vehicle such as an electric vehicle. The vehicle is preferably an autonomous vehicle, a semi-autonomous vehicle, or a remotely-controlled vehicle.

[0016] In variants, the vehicle is a trailer equipped with a trailer coupler adapted to mechanically link to a coupling component of a powered tractor unit. The trailer coupler is preferably a multi-link coupler, which, more preferably, includes or is coupled to a shock-absorbing device.

[0017] According to another aspect, the invention is embodied as an aircraft carrier ground vehicle system. The system comprises one or more aircraft carrier ground vehicles such as described above. In embodiments, the system includes at least two of such vehicles. Each vehicle may for instance be a powered vehicle, which is remotely controlled. In that respect, the system may further include a central control unit that is configured to transmit control signals to each vehicle, and preferably receive sensor signals from each vehicle. The aim is not only to remotely control the rotation of the landing platforms of the vehicles to compensate for an inclination angle of the aircraft on landing, but also to coordinate movements of the vehicles to allow an aircraft to land on the landing platforms of the vehicles as they move along the landing direction of the aircraft, in operation.

[0018] According to a further aspect, the invention is embodied as a method of landing an aircraft, which relies on an aircraft carrier ground vehicle such as described above. According to this method, the vehicle is accelerated, and its speed is adjusted to a landing speed of the aircraft, to bring the vehicle at a landing position of the aircraft. In addition, the landing platform of the vehicle is rotated to match an inclination angle of the aircraft. The aircraft is finally landed on the landing platform as the vehicle moves along the landing direction of the aircraft. As noted above, several vehicles may be used, which can be operated as a swarm.

[0019] According to a final aspect, the invention is embodied as an aircraft take-off method, which relies on an aircraft carrier ground vehicle as described above. According to this method, an aircraft is first loaded and stowed on the landing platform of the vehicle. The vehicle is then accelerated to a take-off speed of the aircraft. The aircraft is released for take-off with the landing platform rotated to a suitable take-off configuration. Again, several vehicles may be relied on.

BRIEF DESCRIPTION OF THE DRAWINGS

[0020] These and other objects, features and advantages of the present invention will become apparent from the following detailed description of illustrative embodiments thereof, which is to be read in connection with the accompanying drawings. The illustrations are for clarity in facili-

tating one skilled in the art in understanding the invention in conjunction with the detailed description. In the drawings:

[0021] FIGS. 1 and 2 are side views of the aircraft carrier vehicle according to embodiments. In FIG. 1, the landing platform is in a horizontal position, while it is slightly pivoted with respect to an axis parallel to the axle direction, which ideally coincides with the pitch axis of the aircraft on landing. The vehicle includes a pneumatic system with pneumatic cylinders supporting the landing platform;

[0022] FIG. 3 is a front view of the aircraft carrier vehicle of FIGS. 1 and 2, further showing an aircraft resting on the landing platform of the vehicle, as of after landing or before take-off, as in embodiments;

[0023] FIG. 4A is a front view of a variant to the vehicle of FIGS. 1 to 3, where the vehicle further includes auxiliary (lateral) dampers, which are supported by additional pneumatic cylinders, as in embodiments. The actuation of the auxiliary dampers is independent from the cylinders actuating the central landing platform;

[0024] FIG. 4B illustrates a variant to FIG. 4A, where the lateral dampers are hinged to the central landing platform, as in embodiments;

[0025] FIG. 5 is a front view of a further variant, in which the landing platform is configured as a Stewart platform, as in embodiments;

[0026] FIG. 6 is a 3D view of a landing platform, illustrating possible rotation axes of a landing platform of a vehicle according to embodiments;

[0027] FIGS. 7 and 8 respectively illustrate the landing and take-off of an aircraft on/from a vehicle with a rotatable landing platform, as in embodiments;

[0028] FIG. 9 shows details of a pneumatic cylinder of a coupling mechanism, which couples the chassis of the vehicle to the landing platform, as in embodiments;

[0029] FIG. 10 is a side view of an aircraft carrier trailer vehicle towed by a powered tractor unit, according to embodiments;

[0030] FIG. 11 is a diagram schematically illustrating components of a system according to embodiments. The system includes several aircraft carrier ground vehicles, as well as a central control unit, which transmits signals to the vehicles to remotely control at least some of the functions of the vehicles, such as the platform rotation, as in embodiments;

[0031] FIG. 12A (front view) and 12B (top view) illustrate how a train of vehicles can be operated for take-off, as in embodiments. FIG. 12C (front view) and 12D (top view) illustrate a different configuration of the vehicles, which may be used on landing the aircraft, as in embodiments;

[0032] FIG. 13 is a graph representing various landing platform configurations and sizes that may be used for take-off and landing of aircrafts of various sizes and weights, as in embodiments;

[0033] FIG. 14 is a 3D view of a deformable (foam) material layer, in which suction cups are embedded. The suction cups are pressurized to grab the aircraft fuselage on landing, or secure the fuselage before take-off, as in embodiments; and

[0034] FIG. 15 is a block diagram schematically illustrating selected components of a control system of a vehicle according to embodiments.

[0035] The accompanying drawings show simplified representations of aircrafts, vehicles, and systems, or parts thereof, as involved in embodiments. Technical features

depicted in the drawings are not necessarily to scale. Similar or functionally similar elements in the figures have been allocated the same numeral references, unless otherwise indicated.

[0036] Vehicles, systems, and methods embodying the present invention will now be described, by way of non-limiting examples.

DETAILED DESCRIPTION OF EMBODIMENTS OF THE INVENTION

[0037] A first aspect of the invention is now described in detail, in reference to FIGS. 1-10. This aspect concerns an aircraft carrier ground vehicle **10**, **10a-d**.

[0038] An “aircraft carrier ground vehicle” is to be interpreted as a vehicle that may notably be railed or wheeled (as assumed in the accompanying drawings), and which is adapted to carry and transport an aircraft. The vehicle may possibly be a trailer **10d** towed by a powered vehicle **30** (i.e., a tractor unit), or a powered vehicle **10**, **10a-c**, such as an electric vehicle, a vehicle equipped with a combustion engine, or a hybrid vehicle. The vehicle may possibly be driven by an operator, autonomous, or remotely controlled (at least partly). The vehicle may also be semi-autonomous, whereby some of its functions are autonomous, while its residual functions can be (possibly remotely) controlled by an operator or a computerized process.

[0039] The vehicle essentially includes a chassis **102**, a landing platform **104**, **104a**, a coupling mechanism **106**, **106c**, and an actuation mechanism **108**, **108a-c**, as illustrated in FIGS. 1-5. Various embodiments are illustrated in the accompanying drawings, where the vehicles shown differ in terms of coupling mechanisms and platform configurations.

[0040] The chassis **102** (i.e., the vehicle frame, or a rolling chassis) is the load-bearing framework of the vehicle; it structurally supports the vehicle and is dimensioned and structured in accordance with the functions to be performed by the vehicle, as described below. The chassis **102** supports the coupling mechanism **106**, **106c**. The actuation mechanism **108**, **108a-c** can be mounted on the chassis, as assumed in the accompanying drawings, or be fully or partly integrated in the chassis. As usual, the chassis **102** may further integrate a motor (should the vehicle be self-powered) and include wheels and transmission.

[0041] The landing platform **104**, **104a** is generally designed to accommodate an aircraft. The landing platform may notably be designed as a continuous landing deck, in which case it can be regarded as a landing bed, as illustrated in FIG. 7. In variants, the platform includes several landing pads, which are mounted on a same platform frame, possibly via separate damping means. The pads may optionally have different relative orientations. The landing pads can further be shaped to optimally fit the lower surface of the fuselage on touch down.

[0042] In embodiments, the vehicle includes distinct landing platforms, which may be individually and independently rotated. In simpler variants, the vehicle **10a**, **10b** includes a main platform (which can be rotated relative to the chassis), as well as auxiliary (lateral) damping pads **104b**, as illustrated in FIGS. 4A and 4B. Note, in the present context, “damping” and “dampers” refer to functions and devices, which make it possible to absorb, alleviate, attenuate, or suppress shocks or vibrations.

[0043] The landing platform **104**, **104a** preferably includes a deformable material layer **105** extending on top of the main platform frame. The deformable material layer **105** may for instance include a foam material. The layer **105** extends on the exposed surface of the platform, as explicitly shown in FIGS. 3-5. The layer **105** deforms to accommodate the shape of the aircraft **20a**, see FIG. 3. The layer may further be designed so as to initially have a conformal shape, e.g., a concave shape to fit the fuselage, a convex shape to fit wings protruding downward, or a more complex shape. Moreover, the deformable layer **105** may advantageously embed suction cups **103**, as depicted in FIG. 14.

[0044] The coupling mechanism **106**, **106c** connects the landing platform **104**, **104a** to the chassis **102**, so as to form a kinematic pair, according to which the landing platform **104**, **104a** is rotatably mounted on the chassis **102**. That is, the kinematic pair imposes constraints on the relative movement of the landing platform and the chassis. The actuation mechanism **108**, **108a-c** is generally designed, and thus adapted, to rotate the landing platform **104**, **104a** relative to the chassis **102**. More precisely, the actuation mechanism causes the landing platform to rotate, to the extent permitted by the coupling mechanism. In other words, the actuation mechanism cooperates with the coupling mechanism to rotate the platform. The aim is to allow an aircraft **20**, **20a** to land on the landing platform **104**, **104a** as the vehicle **10**, **10a-d** moves along the landing direction of the aircraft **20**, **20a**. Rotating the platform makes it possible to compensate for the inclination angle of the aircraft **20**, **20a** and thus to better absorb the shock on landing. So, the present vehicles **10**, **10a-d** can be regarded as remote landing gears with rotatable landing beds for landing aircrafts **20**, **20a**.

[0045] Further comments are in order. The landing platform **104**, **104a** is rotatably mounted on the chassis **102**, meaning that it can pivot around at least one axis, the latter preferably corresponding to the pitch axis of the aircraft as the vehicle aligns its trajectory with that of the aircraft on landing. In embodiments, the platform can rotate around two axes, which correspond to the pitch and the roll axes of the aircraft. The landing platform may further have one or two translational degrees of freedom, whereby the landing platform can be displaced (to a limited extent) parallel to the average plane of the chassis **102** (i.e., parallel to the runway plane, using active translation means) or perpendicular thereto (e.g., using active means to elevate the platform or passive means on damping). So, that the landing platform is rotatable means that that the coupling mechanism **106**, **106c** allows rotation with respect to at least one axis, and preferably two axes, besides possible translational degrees of freedom the platform may have. The fact that the landing platform can be rotated (to the extent permitted by the kinematic pair) allows the aircraft carrier ground vehicle to compensate for (i.e., accommodate for, adapt to) the inclination angle of the aircraft on landing. The inclination angle can be any angle the aircraft has on landing, starting with the pitch angle and, to a lesser extent, the roll angle. The yaw angle is less an issue, since the aircraft will essentially land on its fuselage, which lessens the effect of the slight yaw angle it may have on touch down. Also, the yaw angle can be accommodated for by making the landing bed substantially larger than the aircraft. Conversely, making the landing bed smaller than the fuselage would not necessarily preclude a capture by the platform. So, the landing platform can be made rotatable with respect to only two axes (cor-

responding to the pitch and roll axis). In principle, however, the landing platform and the coupling mechanism may possibly be designed so as for the platform to rotate around all three axes. The pitch, roll, and yaw angles are typically small in practice (e.g., less than 10°, typically between 4 and 7°), such that the landing essentially remains a horizontal-type landing, as opposed to vertical landing approaches, notwithstanding the ability of the platform to rotate. Note, the apparent pitch angle in FIG. 7 is purposely exaggerated, in the interest of depiction.

[0046] The aircraft **20**, **20a** may notably be a spacecraft designed to land as a glider. However, the spacecraft may also be self-powered and may possibly use propulsion on landing. In practice, the aircraft may weigh up to several (even hundreds) of tons. Thus, the dimensions and mechanical resistance properties of the platforms, the coupling mechanism, the chassis, and the actuation mechanism, must be commensurate with characteristics dimensions, mass, mechanical resistance, vertical speed, etc., of the aircraft on landing.

[0047] The coupling mechanism mechanically couples the platform with the chassis. Various coupling devices may potentially be involved, e.g., pneumatic cylinders, hydraulic cylinders, and/or hydropneumatic devices. In particular, the coupling mechanism may combine different types of systems, such as pneumatic and hydraulic systems. For instance, the coupling system may involve one or more pneumatic air-oil hydraulic shock absorbers such as oleo struts, complemented by further means to rotate the platform (e.g., pneumatic cylinders). In simple embodiments, however, the coupling mechanism relies on pneumatic cylinders only, as assumed in the accompanying drawings.

[0048] For instance, the coupling mechanism may possibly be designed so as for the platform to have a hexapod-or Stewart platform-like configuration, as in FIG. 5. Care should be taken for the coupling design to prevent self-locking. To that aim, the coupling mechanism may involve various types of joints, such as ball-and-socket joints **109**, planar joints, pin joints, rack and pinions, etc. Thus, the coupling mechanism may enable various attitude controls, which can be exploited to best receive the aircraft on landing. Because the coupling mechanism **106**, **106c** mechanically connects the landing platform **104**, **104a** with the chassis **102** in a way that constraints the possible degrees of freedom of the platform, the landing platform **104**, **104a** and the chassis **102** form a kinematic pair.

[0049] Besides the coupling mechanism, latching mechanisms may be involved, such as a hard robotic mechanical latching, a soft robotic suction cup latching, a magnetic attraction mechanism, or a hybrid mechanism. The aim is to be able to grasp the aircraft as the latter is landing, e.g., to prevent the aircraft from bouncing. In addition, the platform may possibly include fasteners to secure the aircraft on the platform once it has landed. Once the aircraft is secured on the platform, the kinematic pair formed by the platform and the chassis constraints the relative movement of, on the one hand, the landing platform **104**, **104a** and the aircraft **20**, **20a** and, on the other hand, the chassis **102**.

[0050] The actuation mechanism **108**, **108a-c** is preferably mounted on or (at least partly) integrated in the chassis **102**, be it to lower the gravity centre of the vehicle. In the accompanying drawings, the actuation mechanism is assumed to be mounted on the chassis. In variants, the actuation mechanism is mounted on the platform, hence

offering a different weight ratio between the platform and the chassis. The actuation mechanism **108**, **108a-c** cooperates with the coupling mechanism, at least indirectly. E.g., where the coupling mechanism consists of pneumatic cylinders, the actuation mechanism includes pressurization means that pressurizes a working fluid (e.g., air or nitrogen) in the rodless cavity under the piston head and/or in the piston rod cavity to actuate the piston rod. I.e., the actuation mechanism cooperates with the pneumatic cylinders, mechanically, to rotate the platform. In variants, the actuation mechanism may operate independently of the coupling mechanism. E.g., electromagnetic means may be used to rotate the platform. In that case, the actuation mechanism only indirectly cooperates with the coupling mechanism, although actuating the platform still impacts the kinematic pair and thus the configuration of the coupling. In further variants, the actuation mechanism and the coupling mechanism may form part of a very same mechanism (e.g., a mechanical linkage).

[0051] Both the landing platform and the chassis may advantageously be aerodynamically profiled and/or include aerodynamic structures. In particular, the vehicle may include active aerodynamic features, such as a motorized front spoiler **120**, as assumed in FIGS. 1 and 2. The vehicle may further be equipped with passive aerodynamic features (not shown), such as an air dam, a rear spoiler, etc., to control the aerodynamic forces acting upon the system at high speeds (e.g., reduce drag during acceleration and increase it during deceleration, change vehicle centre of pressure, increase or decrease downforce). The tractor unit **30** may similarly be aerodynamically profiled, should the aircraft carrier vehicle be embodied as a trailer.

[0052] The present solution has several advantages. The invention proposes an aircraft carrier ground vehicle equipped with a pivotable landing platform, which can be regarded as a remote landing gear. Thus, the aircraft does not need to be permanently equipped with a landing gear. This makes it possible to reduce the aircraft mass or conversely increase the payload of and volume available in the aircraft. Moreover, this allows the structural efficiency of the aircraft to be improved. In particular, the absence of landing gear makes it possible to relax structural reinforcement constraints that normally apply to the load bearing structure to accommodate the landing gear. However, the load bearing structure still need to be designed to allow landing on the fuselage.

[0053] In addition, the proposed approach allows an easier maintenance of the aircraft. Moreover, this approach may be used to achieve smaller landing and take-off distances, e.g., by leveraging propulsion and braking capabilities of the vehicle (or a tractor unit thereof), in addition to those of the aircraft.

[0054] The present aircraft carrier ground vehicles may be used for landing only, should the aircraft use other take-off means, such as a launching ramp for vertical take-off (possibly assisted by boosters), another type of aircraft carrier ground vehicle (e.g., railed instead of being wheeled), or an ejectable landing gear. In variants, the same vehicle may also be used for taxiing and take-off, as in embodiments discussed below.

[0055] For completeness, the present approach is fairly easily scalable. It can notably be applied to aircrafts weighing tens of kilograms to hundreds of tons and ranging from low-speed aircrafts such as high-altitude long endurance

aircrafts, including drones to high-speed vehicles such as hyperplanes and space planes, as illustrated in FIG. 13. All this is now described in detail, in reference to particular embodiments of the invention. To start with, the coupling mechanism **106**, **106c** shall typically be designed to allow the average plane of the landing platform **104**, **104a** to pivot by at most a first angle around an axis that is parallel to the axle direction of the vehicle **10**, **10a-d**. The design of the coupling mechanism (e.g., relying on pneumatic cylinders) typically limits the rotation angle(s). In the present case, the coupling mechanism may be designed so as to restrict the maximal rotation angle(s) to less than 10° , or even less than 8° or 7° . The axle direction corresponds to axis y in the accompanying drawings.

[0056] In other words, the coupling mechanism **106**, **106c** allows the landing platform to pivot with respect to the average plane of the chassis **102**, around the axle direction, which ideally coincides with the pitch axis of the aircraft on landing, see FIG. 6. The average plane of the chassis is parallel to plane (x, y) in the drawings. The axle direction is parallel to the plane (x, y) and orthogonal to the longitudinal direction x of the vehicle. So, the axle direction is parallel to the runway plane and orthogonal to the landing direction, at least under ideal landing conditions. Pivoting the platform around the pitch axis makes it possible to accommodate for the aircraft pitch angle on landing.

[0057] Similarly, the coupling mechanism **106**, **106c** may be designed to allow the average plane of the landing platform **104**, **104a** to pivot by at most a second angle around the longitudinal direction x. That is, the coupling mechanism **106**, **106c** further allows the landing platform **104**, **104a** to pivot with respect to an axis that ideally corresponds to the roll axis, which extends parallel to the longitudinal direction of the vehicle. This makes it possible to accommodate for the roll angle of the aircraft **20**, **20a** on landing. The maximal rotation angle permitted around the roll axis can be substantially less than the first angle (pitch angle), in practice, consistently with the fact that the roll angle of the aircraft will likely be less than its pitch angle on landing. In that respect, the coupling mechanism and the actuation mechanism may possibly be designed so as to restrict the maximal rotation angle around the roll axis, notably for it to be less than the maximal rotation angle around the pitch axis, e.g., less than 8° or 7° , or less.

[0058] As noted earlier, the coupling mechanism of the vehicle **10**, **10a-d** may notably involve pneumatic cylinders, as assumed in FIGS. 1-10. More precisely, the coupling mechanism **106**, **106c** may include a pneumatic suspension system, which involves at least two pneumatic cylinders **106**, **106c**, where the cylinders connect the chassis **102** to the landing platform **104**, **104a**. Two pneumatic cylinders may suffice to pivot the platform around one axis, in principle.

[0059] In practice, however, the coupling mechanism may include more than two pneumatic cylinders **106**, **106c**, e.g., four pneumatic cylinders, as in FIGS. 1-3. The landing platform typically forms a quadrangle (in top view), such that each cylinder may connect a respective corner of the platform. Each piston may either connect to the chassis **102** or the platform **104** (as assumed in the accompanying drawings), while the respective cylinder tube (barrel) connects to the companion part (the platform or the chassis). Such connections typically involve joints, as discussed later in detail. Additional pneumatic cylinders **106a**, **106b** may

further be involved, e.g., to support auxiliary damping pads **104b**, as in FIGS. 4A and 4B.

[0060] All such pneumatic cylinders **106**, **106c** can be actuated by an actuation mechanism **108**, **108a-c**, notably to rotate the landing platform **104**, **104a** relative to the chassis **102**. Actuating the pistons of the pneumatic cylinders via the actuation mechanism allows the landing platform to be pivoted around one or two rotation axes. The pneumatic cylinders are arranged about peripheral portions of the platform to achieve the desired rotation(s), e.g., on opposite sides of the platform. In principle, two pneumatic cylinders can be arranged on opposite sides of the platform to pivot the latter around one axis, e.g., the pitch or roll axis. Adding two opposite cylinders on the remaining sides further allows the platform to be rotated around a second axis.

[0061] Thus, the pneumatic suspension system may possibly include four pneumatic cylinders **106**, **106c**. Adding further pneumatic cylinders will provide additional power, stability, and damping. For example, the pneumatic suspension system may include six pneumatic cylinders **106c**, as shown in FIG. 5, where the platform **104** and the pneumatic suspension system **106c** are jointly configured as a Stewart platform (also known as Gough-Stewart platform).

[0062] FIGS. 1-5 assume the use of pneumatic cylinders. In variants, hydropneumatics devices (e.g., similar to oleo struts) can be devised, which combine gas (e.g., nitrogen) and liquids (e.g., hydraulic oil) to absorb the impact energy and dissipate it, respectively. Modulating the gas pressure and/or liquid quantity allows platform attitudes such as pivot angles to be controlled. In other variants, the landing platform can be primarily coupled to the chassis via a shaft, a ball-and-socket joint, or any other suitable type of joint allowing rotation (not shown). For example, use can be made of a spherical ball-and-socket joint, including a spherical ball (or a spherical shape) at the centre of the landing platform, where the spherical ball (or shape) is received in a complementary shaped part in the chassis (e.g., in the vicinity of its centre), thanks to which the landing platform can rotate relative to the chassis. In that case, auxiliary coupling means such as pneumatic cylinders can be used to rotate the landing platform; one or two pneumatic cylinders may suffice. Electromagnetic actuation means may be used too. However, the primary coupler does not ensure any damping, such that an additional damping mechanism is needed. One may, in principle, leverage the damping (e.g., suspension) mechanism of the chassis itself. Preferably, however, auxiliary damping means are used. For example, the complementary shaped part of the chassis receiving the spherical part of the joint may be mounted on a pneumatic suspension that is distinct from the damping mechanism of the vehicle.

[0063] Preferred, however, is to rely on a pneumatic system involving at least four pneumatic cylinders, which are not only used to rotate the platform but also serve as dampers on touch down. In that respect, the actuation mechanism **108**, **108a-c** may advantageously include pressurization means connected to each cavity (the rodless cavity and the rod cavity) of each pneumatic cylinder **106**, **106c**, with a view to operating each of the pneumatic cylinders **106** independently, **106c** as double-acting cylinders, as assumed in FIG. 9. In this example, two valves **112**, **114** are provided on the cylinder part **1064**, which connect to pressurization means of the actuation mechanism **108** via respective pneumatic hoses **111**, **113**, whereby pressurized

air (or another working fluid) can be injected on both sides of the piston rod **1061** to act thereon and rotate the platform **104**. The cylinder shown in FIG. 9 is preferably operated as follows. A pressurized fluid is injected via controlled valves **112** and/or **114** to actuate the piston rod and rotate the platform, for it to reach a desired landing configuration. The valves may automatically close after each pressure adjustment. In variants, pressure is maintained in each cylinder cavity until the aircraft lands on the platform. Upon touching the platform, the aircraft compresses the piston, which causes to compress the fluid in the rodless cavity, hence dampening the shock.

[0064] More sophisticated types of cylinder system can be contemplated in variants. For example, the cylinder may be designed as an oleo strut, whereby a metering pin protrudes in an inner oil chamber, to compress gas on top and dampen the rebound movement. The following description, however, assumes that use is made of pneumatic cylinders, which allow fairly simple designs and fabrication.

[0065] As further seen in FIG. 9, the coupling mechanism **106** may include ball-and-socket joints **109** linking the piston rods **106** of respective pneumatic cylinders **106** to the landing platform **104**, to enable the desired rotations and prevent self-locking. Note, the ball-and-socket joints **109** may possibly be arranged at the top of the piston rod (as in FIG. 9), or at the bottom of the cylinder tubes (barrels) **1064**, or both. However, distinct linking mechanisms can be used at the top and bottom of the cylinders **106**.

[0066] For instance, the coupling mechanism **106** may further comprise revolute joints **116**, **118** (also called pin joints or hinge joints) linking respective cylinders **106** to the chassis **102**, as further illustrated in FIG. 9. In this example, the revolute joints are below the cylinder tubes. In simpler variants, the cylinders may be rigidly fixed to the chassis or fixed via a spring joint, to further damp on touch down, for example. As noted earlier, the chassis **102** typically includes, or is coupled to, a damping mechanism, such as a car-like air suspension mechanism or a leaf-spring suspension. Several damping mechanisms may thus be cascaded, to better absorb the shock on touch down.

[0067] As further illustrated in FIG. 9, the cylinder tube **1064** of each cylinder **106** can advantageously be set in thermal communication with a heat sink **102**, e.g., via a metallic part **115**. The heat sink may notably be the chassis **102** itself or form part thereof. The aim is to rapidly absorb heat produced upon rapidly compressing the pneumatic cylinders on touch down.

[0068] As depicted in FIGS. 3-5 and 14, the landing platform **104**, **104a** preferably includes a deformable material layer **105**, preferably made from a foam material, which extends on the exposed surface of the landing platform **104**, **104a**. Auxiliary dampers **104b** may include such a deformable layer too. This layer **105** shall preferably deform elastically, so as to be reusable. The deformable layer **105** provides an additional damping means. E.g., a foam mattress can be provided to complement the damping actions of the damping system of the vehicle and the pneumatic cylinders **106**, **106c**, whereby some of the kinetic energy transferred on landing is absorbed by the deformable layer.

[0069] As evoked above, the vehicle **10**, **10a-d** may optionally include one or more auxiliary dampers **107**, **107b**, each including a landing pad **104b**, which is preferably covered with a deformable layer, similar to layer **105** on the main landing platform **104a**. This way, the aircraft **20**, **20a**

can land on both the landing platform **104a** and the auxiliary dampers **107**, **107b**. The additional dampers **107** can be arranged on any side of the main landing platform **104a** (e.g., in front of or behind the main platform **104a**, or on a lateral side), so as to best accommodate for the underside of the aircraft, see FIGS. 4A and 4B. Such dampers may for instance be mounted on two or more pneumatic cylinders **106a**, as in FIG. 4A, or laterally hinged to the central platform **104a**, and damped thanks to one or more pivotable cylinder pistons **106b**, as assumed in FIG. 4B. The design of the actuation unit **108b** can easily be adapted to connect to and pressurize the additional pneumatic cylinders **106a**, **106b**. On landing, the pads **104b** move (if necessary, asymmetrically) to best accommodate the aircraft position, which may likely be slightly off-nominal.

[0070] As seen in FIG. 14, the vehicle may further include an array of suction cups **103** that are embedded in the deformable material layer **105**. The suction cups **103** are open on the exposed surface of the layer **105**. The suction cups **103** are connected to a pressurization system **110**, which is preferably distinct from the pressurization means **108**, **108a-c** used to actuate the pneumatic cylinders, to suitably decouple operations. The pressurization system **110** may be arranged under the platform **104** or laterally. The pressurization system **110** is used to create a depression in the cups, allowing the cups to grip the aircraft on landing or secure it before take-off. In practice, one will attempt to achieve a pressure in the cups that is significantly lower than the ambient atmospheric pressure (e.g., less than 500 mbar or, even, less than 100 mbar, depending on the size and mass of the aircraft, and the number and size of the cups).

[0071] The suction cups **103** can be pressurized to grab the fuselage on landing, notably to prevent bouncing (normal force) and sliding of the aircraft during deceleration (shear force). They can also maintain the fuselage when taxiing and before take-off (shear force). The suction cups **103** can be fully embedded in the foam bed, see FIG. 14, such that their apertures are flush (or essentially flush) with the top surface of the deformable material layer **105**. Still, the apertures of the cups may slightly protrude (e.g., by a few millimetres) above the top surface of the deformable material layer **105**. That the layer **105** deforms on landing allows the upper plane of the suction cups to better adapt to the local surface geometry of the aircraft and, thus, to maintain a good grasp. Conversely, the pressurization system **110** may possibly be actuated to provoke a positive pressure surge, to help the aircraft to separate from the platform on take-off.

[0072] Instead of, or in addition to pressurized suction cups (which are a soft robotic latching means), other latching means may be involved, which are adapted for gripping a structural part of the aircraft **20**, **20a** on landing. Examples of suitable latching means include a hard robotic mechanical latching (i.e., mechanical gripping means) and a magnetic attraction mechanism. The structural part at issue may notably be the skin of the aircraft (e.g., the skin of the fuselage or wings) or another internal structural element, such as spars or stringers exposed through the opening of a hatch in the aircraft skin.

[0073] In addition, the vehicle may use one or more brake systems, including aerodynamic braking elements or devices (e.g., airbrake, spoiler, parachute), disk-braking at high speeds (e.g., hydro-dynamic braking, eddy current braking) and low speeds (e.g., carbon ceramic brake disks and pads). Moreover, the vehicle may optionally include further equip-

ment, such as a compact arresting system and/or a compact catapulting system, as well as active aerodynamic surfaces for controlling vehicle dynamics.

[0074] As noted earlier, the present vehicles may be autonomous, semi-autonomous, or remotely controlled (e.g., by an automated process), whereby some or all their functions may be automatically performed. To that aim, the present vehicles may include an advanced vehicle control system 360. A high-level functional diagram of such a control system 360 is shown in FIG. 15. The control system 360 may notably include one or more control units 1085 connected to the actuation mechanism 108 to control a rotation of the landing platform 104 relative to the chassis 102. In addition, the system 360 may include receivers 302 coupled to the control unit(s) 1085. Such receivers 302 are configured to receive instruction signals, e.g., from a remote-control unit 11 (e.g., forming part of a checkpoint or a control tower, see FIG. 11), which they can then forward to a computerized unit 202, for the latter to forward corresponding control signals to the control units 1085. Intermediate processing may be performed at the computerized unit 202, if necessary. The receivers 302 may in fact be transceivers, or the vehicle may further include transmitters 304, whereby the vehicle control system 360 can transmit sensor readings to the external control unit 11, with a view to remotely controlling the in-vehicle actuation mechanism 108 and, optionally the vehicle speed and direction too. In fact, various telemetry values may be remotely transmitted, including tank and cylinder pressures, battery system voltages and current draw, GPS position, break temperature, impact force etc.

[0075] In the example of FIG. 15, sensors 310 are provided, which are configured to detect events or changes in the environment of the vehicle. The sensors accordingly produce signals, which are transmitted to a receiving unit 320, where the signals may be pre-processed, if needed. The computerized system 202 is connected to the receiving unit 320, so as to receive signals from the unit 320. Similarly, the computerized system 202 receives and processes signals from the receiver 302. Signals received from the sensor receiving unit 320 and/or the receiver 302 may possibly require further processing, in order to be exploited by software executing at the central processing unit (CPU) 230 of the computerized system 202. In operation, computerized methods (typically in the form of software) are loaded in the main memory 250 of the system 202 to cause the processing means 230 to produce real-time signals to actuate the actuation mechanism 108 and other units of the vehicle (notably a motion control unit), based on the signals received from the receivers 302 and the receiving unit 320. In particular, the computerized system 202 may be con-

nected to a motion control unit 360 and instruct the latter to operate the vehicle in a partly or fully autonomous manner, or automated manner. As usual, the system 202 may include memory controllers 240, a long-term storage means 255, a system bus 270, amongst other components. All required components for enabling such an autonomous, semi-autonomous, or remotely controlled vehicle, are known per se.

[0076] Computerized devices can be suitably designed for implementing embodiments of the present invention. In that respect, it can be appreciated that some of the methods described herein are at least partly non-interactive, i.e., automated. Automated parts of such methods will typically be implemented as a combination of hardware and software.

[0077] In the examples of FIGS. 7 and 8, the aircraft carrier ground vehicle 10 is a powered vehicle, for example an electric vehicle. However, the aircraft carrier vehicle may also be a trailer 10d, equipped with a trailer coupler 125, which mechanically links to a coupling component of a powered tractor unit 30, as illustrated in FIG. 10. The trailer may also be powered, hence supplementing the tractor unit to achieve the required horsepower. The trailer coupler 125 is preferably a multi-link coupler, i.e., with two or three attachment points. A multi-link coupler is used to increase the trailer stability when landing the aircraft 20. The coupler, whether multi-link or not, may further include or be coupled to a shock-absorbing device (not shown). Like the powered vehicles discussed earlier, the trailer is preferably multi-axled to maintain stability on landing and take-off, as seen in FIG. 10. The axle position can be optimized to favour stability, by notably ensuring that the centre of gravity of the system (vehicle or vehicle +aircraft) is well in the middle of the axles.

[0078] The following tables illustrate characteristics of examples of aircraft configurations and aircraft carrier vehicle systems according to embodiments.

TABLE 1

Characteristics of representative landing aircrafts				
Class	Medium drone	Medium airliner	Large airliner	Suborbital spaceplane
Maximum take-off weight (kg)	4760	78245	356000	325000
Maximum landing weight (kg)	4284	65317	255800	55000
Fuselage length (m)	11	39.5	70.51	83.13
Acceleration at take-off (m/s ²)	2.3	1.96	2.3	2.3
Deceleration at landing (m/s ²)	4.6	4.6	4.6	4.6

TABLE 2

Characteristics of a suitable aircraft carrier vehicle for a medium airliner (78 tons)				
Component	Sub-component	Material	Quantity	Mass (Kg)
Coupling system	Suction cups	Nitrile-butadiene-rubber (NBR) (density 1000 kg/m ³)	4330	139
	Foam	Elastomer (density 1100 kg/m ³)	1	606
	Base plate	Aluminium alloy 7075 sandwich structure (300 kg/m ³)	1	1575

TABLE 2-continued

Characteristics of a suitable aircraft carrier vehicle for a medium airliner (78 tons)				
Component	Sub-component	Material	Quantity	Mass (Kg)
Platform and actuation system	Ball joints	Steel alloy	4	2
	Oleo-pneumatic struts	Steel alloys/hydraulic fluid/air	4	2286
	Vacuum tank	Stainless steel (7500 kg/m ³)	1	113
	Pump (7.5 HP)	Steel and iron alloys	1	80
	Universal joints	Steel alloy	4	2
Base	Trailer base	Steel alloys	1	8000
	Wheels	Polymers	18	810

[0079] In the example of Table 2, the total mass of the aircraft carrier system is approximately equal to 13.61 tons. It involves a single vehicle with a single unpowered axle (two unpowered wheels) and two tandem axles powered by four electromotors each (eight powered wheels), resulting in 10x8 tractions.

(two unpowered wheels), two tandem axles powered by four electromotors each (eight powered 10 wheels) and one unpowered lift axle (2 unpowered wheels), resulting in 12x8 traction.

[0081] A number of variants can be contemplated. E.g., the number of aircraft carrier ground vehicles can be

TABLE 3

Characteristics of an aircraft carrier vehicle used for a large airliner (356 tons)				
Component	Sub-Component	Material	Quantity	Mass (Kg) L × W × H (m)
Coupling system	Suction cups	Nitrile-butadiene-rubber (NBR) (density 1000 kg/m ³)	16956	543 0.05 × 0.05 × 0.05
	Foam	Elastomer (density 1100 kg/m ³)	1	2372 21.19 × 2.50 × 0.05
	Base plate	Aluminium alloy 7075 sandwich structure (300 kg/m ³)	1	1575 21.69 × 2.50 × 0.15
Platform and actuation system	Ball joints	Steel alloy	4	2 0.05 × 0.56 × 0.05
	Oleo-pneumatic struts	Steel alloys/hydraulic fluid/air	4	2800 1.00 × 0.05 × 0.05
	Vacuum tank	Stainless steel (7500 kg/m ³)	1	113 0.4 × 0.4 × 6.0
	Pump (7.5 HP)	Steel and iron alloys	1	80 0.465 × 0.64 × 0.33
	Universal joints	Steel alloy	4	2 0.07 × 0.07 × 0.14
Base	Trailer base	Steel alloys	1	8000 13.6 × 2.5 × 1.0
	Wheels	Polymers	18	810 1.06 × 0.45 × 0.45

[0080] In the example of Table 3, the total mass of the aircraft carrier system is approximately equal to 65.2 tons. The system involves four identical aircraft carrier vehicles, each having a mass of approximately 16.3 tons. Each aircraft carrier vehicle includes a single unpowered steering axle

changed. For example, a single vehicle as specified in Table 3 can be used to perform the same mission as the example vehicle in Table 2. Conversely, several such vehicles can be used to perform the same mission as the example vehicle in Table 4 below.

TABLE 4

Characteristics of a suitable aircraft carrier vehicle for suborbital spaceplane (325 tons take-off mass and 55 tons landing mass)				
Component	Sub-Component	Material	Quantity	Mass (Kg) L × W × H (m)
Coupling system	Suction cups	Nitrile-butadiene-rubber (NBR) (density 1000 kg/m ³)	3646	117 0.05 × 0.05 × 0.05
	Foam	Elastomer (density 1100kg/m ³)	1	510 4.56 × 2.50 × 0.05

TABLE 4-continued

Characteristics of a suitable aircraft carrier vehicle for suborbital spaceplane (325 tons take-off mass and 55 tons landing mass)					
Component	Sub-Component	Material	Quantity	Mass (Kg)	L × W × H (m)
Platform and actuation system	Base plate	Aluminium alloy 7075 sandwich structure (300 kg/m ³)	1	1575	5.06 × 2.50 × 0.15
	Ball joints	Steel alloy	4	2	0.05 × 0.56 × 0.05
	Oleo-pneumatic struts	Steel alloys/hydraulic fluid/air	4	1925	1.00 × 0.05 × 0.05
	Vacuum tank	Stainless steel (7500 kg/m ³)	1	113	0.4 × 0.4 × 6.0
	Pump (7.5 HP)	Steel and iron alloys	1	80	0.46 × 0.64 × 0.33
	Universal joints	Steel alloy	4	2	0.07 × 0.07 × 0.14
Base	Trailer base	Steel alloys	1	8000	13.60 × 2.50 × 1.00
	Wheels	Polymers	18	810	1.06 × 0.45 × 0.45

[0082] In the example of Table 4, the total mass of the aircraft carrier system is approximately equal to 13.13 tons; it involves a single vehicle for landing a returning aircraft (due to the significantly lower landing mass). Four vehicles are used for take-off and landing a fuelled-up spaceplane in a mission-abort scenario.

[0083] A preferred mass ratio of the aircraft (whether at take-off or landing) to the system of aircraft carrier vehicles is preferably between 4 and 7, more preferably between 5 and 8, to ensure a satisfactory mechanical behaviour of the overall system (consisting of the aircraft and the aircraft carrier vehicle system).

[0084] Referring to FIGS. 11-13, another aspect of the invention is now described, which concerns an aircraft carrier ground vehicle system 1. The system comprises at least one, and preferably more than one aircraft carrier ground vehicles 10, 10a-d such as described in reference to the first aspect of the invention. The system 1 may for instance comprise several powered ground vehicles 10, each capable of accelerating to typical landing speeds of an aircraft, through built-in propulsion. In variants, some of the vehicles may be unpowered or sub-powered vehicles, which are towed by one or more tractor units in a tractor-trailer(s) configuration. The ground vehicles act as pivotable platforms.

[0085] The vehicles are preferably unmanned, remotely controlled vehicles. Their platforms may advantageously be remotely controlled thanks to computerized methods as noted above. To that aim, the system 1 may include a central control unit 11 that is configured to transmit control signals to each vehicles 10 via an antenna 12, as illustrated in FIG. 11. As noted earlier too, the central control unit 11 may possibly receive sensor signals from each or some of the vehicles, so as to remotely instruct to suitably rotate the landing platforms 104 of the vehicles to compensate for the inclination angle(s) of the aircraft 20. Further instructions may be sent to the vehicles to coordinate movements of the vehicles, to allow the aircraft 20 to land on the multiple landing platforms of the vehicles.

[0086] In such embodiments, the vehicles 10 are remotely controlled and behave as a swarm, which, as a whole, can be operated to ensure a safe landing of the aircraft 20. There are multiple advantages to operating the vehicles in a swarming

configuration. First, the vehicle swarm can be operated to provide multiple landing points, as illustrated in FIGS. 12C, 12D, and 13, notably for heavy aircrafts (typically of more than 20 tons). There, the multiple vehicles involved will typically distribute (laterally and/or longitudinally) under the aircraft to best receive it. Conversely, on take-off, a train of vehicles (whether mechanically connected or not) can be used to help propel the aircraft. E.g., one or more vehicles may be used to push or pull the vehicle that initially supports the aircraft, to ensure redundancy on take-off, as illustrated in FIGS. 12A and 12B. Having a system of swarmed vehicles can be extremely helpful, notably in emergency situations. For example, if the aircraft must operate an emergency landing short after take-off, the vehicle swarm, which is already deployed on the runway, can quickly reorganize to adopt a lateral configuration and land the aircraft.

[0087] Final aspects of the invention concerns landing and take-off methods, which rely on aircraft carrier ground vehicles 10, or systems 1 of such vehicles, as described above. Such methods aim at operating an aircraft that is free of a landing gear. Several features of such methods have already been described above, be it implicitly, in reference to other aspects of the invention. Such methods are only succinctly described in the following.

[0088] FIGS. 7, 12C, and 12D illustrate a method of landing an aircraft 20, using an aircraft carrier ground vehicle 10. In short, the vehicle is accelerated, and its speed is adjusted to the landing speed of the aircraft 20, to bring the vehicle at a landing position of the aircraft 20 (i.e., an expected touch down position), e.g., using a remote control or autonomous driving method. In addition, the landing platform 104 of the vehicle is rotated to match the inclination angle of the aircraft 20. Note, the landing platform may initially be rotated, prior to or while accelerating the vehicle. Moreover, the rotation angle(s) may possibly be adjusted, dynamically, during the approach phase. The landing platform may typically be pivoted with respect to the pitch axis, and/or the roll axis, to compensate for crosswind, if necessary. Similarly, the vehicle speed may be repeatedly adjusted to best meet the rendezvous point. The aircraft 20 eventually lands on the landing platform as the vehicle moves along the landing direction of the aircraft 20.

[0089] FIGS. 8, 12A, and 12B, illustrate an aircraft take-off method, wherein an aircraft 20 is first loaded to and stowed on the landing platform of the vehicle 10, prior to accelerating the vehicle to a take-off speed of the aircraft 20. Once the combined system (vehicle and aircraft) has achieved the desired take-off speed, the aircraft 20 is released for take-off. The landing platform may be slightly rotated around to achieve a take-off configuration. The platform may for instance be rotated prior to or while accelerating the carrier vehicle, e.g., pivoted with respect to the pitch axis, and/or the roll axis.

[0090] While the present invention has been described with reference to a limited number of embodiments, variants, and the accompanying drawings, it will be understood by those skilled in the art that various changes may be made, and equivalents may be substituted without departing from the scope of the present invention. In particular, a feature (device-like or method-like) recited in a given embodiment, variant or shown in a drawing may be combined with or replace another feature in another embodiment, variant, or drawing, without departing from the scope of the present invention. Various combinations of the features described in respect of any of the above embodiments or variants may accordingly be contemplated, that remain within the scope of the appended claims. In addition, many minor modifications may be made to adapt a particular situation or material to the teachings of the present invention without departing from its scope. Therefore, it is intended that the present invention is not limited to the particular embodiments disclosed, but that the present invention will include all embodiments falling within the scope of the appended claims. In addition, many other variants than explicitly touched above can be contemplated. For example, other types of coupling mechanisms and joints may be involved.

REFERENCE LIST

System

- [0091] Aircraft carrier ground vehicle system 1
- [0092] Central control unit 11
- [0093] Transmission/reception antenna 12

Vehicles and Equipment

- [0094] Self-powered vehicle 10, 10a-c
- [0095] Aircraft carrier ground vehicle 10, 10a-d
- [0096] Trailer vehicle 10d
- [0097] Aircraft 20, 20a
- [0098] Powered tractor unit 30
- [0099] Chassis/heat sink 102
- [0100] Front spoiler 120
- [0101] Trailer coupler 125

Coupling Means and Joints

- [0102] Piston rod 1061
- [0103] Piston head 1062
- [0104] Cylinder tube/barrel 1064
- [0105] Coupling mechanism (pneumatic cylinders) 106, 106c
- [0106] Auxiliary dampers 107, 107b
- [0107] Revolute joint (pin/hinge joint) 116, 118
- [0108] Ball-and-socket joints 109
- [0109] Thermal communication element 115

Actuation Means

- [0110] Actuation mechanism 108, 108a-c
- [0111] Hoses 111, 113
- [0112] Controlled valves 112, 114

Platform

- [0113] Landing platform 104, 104a
- [0114] Auxiliary damper pads 104b
- [0115] Suction cups 103
- [0116] Deformable material layer (foam material layer) 105
- [0117] Pressurization system 110

Vehicle Control System 300

- [0118] Receiver 302
- [0119] Transmitter 304
- [0120] Sensors 310
- [0121] Sensor receiving unit 320
- [0122] Vehicle motion control unit 360
- [0123] Actuation mechanism control units 1085
- [0124] Computerized unit 202

Computerized Unit Modules and Components

- [0125] CPU 230
- [0126] Memory controller(s) 240
- [0127] Memory 250
- [0128] Long-term storage 255
- [0129] System bus 270

1. An aircraft carrier ground vehicle, comprising:
a chassis;

a landing platform designed to accommodate an aircraft;
a coupling mechanism, which connects the landing platform to the chassis to form a kinematic pair, in which the landing platform is rotatably mounted on the chassis; and;

an actuation mechanism adapted to rotate the landing platform relative to the chassis, to allow an aircraft to land on the landing platform as the vehicle moves along a landing direction of the aircraft and compensate for an inclination angle of the aircraft on landing, in operation; and

latching means adapted for gripping a structural part of the on landing, wherein the latching means includes a soft robotic suction cup latching mechanism.

2. The aircraft carrier ground vehicle according to claim 1, wherein

the coupling mechanism is designed to allow an average plane of the landing platform to pivot by at most a first angle around an axis that is parallel to an axle direction of the vehicle.

3. The aircraft carrier ground vehicle according to claim 2, wherein

the coupling mechanism is further designed to allow the average plane of the landing platform to pivot by at most a second angle around a longitudinal direction of the vehicle.

4. The aircraft carrier ground vehicle according to any one of claim 1, wherein

the coupling mechanism includes a pneumatic suspension system having at least two pneumatic cylinders connecting the chassis to the landing platform, and

- the pneumatic cylinders are actuatable by the actuation mechanism to rotate the landing platform-relative to the chassis, in operation.
5. The aircraft carrier ground vehicle according to claim 4, wherein
- the actuation mechanism includes pressurization means connected to each of the pneumatic cylinders, so as to independently operate the pneumatic cylinders as double-acting cylinders.
6. The aircraft carrier ground vehicle according to claim 4, wherein
- the coupling mechanism further comprises ball-and-socket joints linking piston rods of respective ones of the pneumatic cylinders-to the landing platform.
7. The aircraft carrier ground vehicle according to claim 4, wherein
- the coupling mechanism further comprises revolute joints linking respective ones of the pneumatic cylinders to the chassis.
8. The aircraft carrier ground vehicle according to claim 4, wherein
- a cylinder tube of each of the pneumatic cylinders is in thermal communication with a heat sink, the latter forming part of the chassis.
9. The aircraft carrier ground vehicle according to claim 1, wherein
- the landing platform further includes a deformable material layer, extending on an exposed surface of the landing platform.
10. The aircraft carrier ground vehicle according to claim 9, wherein
- the vehicle further includes a pressurization system,,
- the soft robotic suction cup latching mechanism comprises an array of suction cups, which are at least partly embedded in the deformable material layer,
- the suction cups are open on an exposed surface of the deformable material layer, and
- the suction cups are connected to said pressurization system.
11. The aircraft carrier ground vehicle according to claim 1, wherein
- the vehicle further includes one or more auxiliary dampers, each including a landing pad, so as to allow the aircraft to land on both the landing platform and the one or more auxiliary dampers as the vehicle moves along the landing direction of the aircraft, in operation.
12. (canceled)
13. The aircraft carrier ground vehicle according to claim 1, wherein the vehicle further includes
- a vehicle control system, the latter comprising
- one or more control units coupled to the actuation mechanism to control a rotation of the landing platform relative to the chassis, and
- one or more receivers coupled to the one or more control units, the receivers configured to receive instruction signals for the vehicle control system to forward corresponding control signals to the one or more control units.
14. The aircraft carrier ground vehicle according to claim 1, wherein
- the vehicle is a powered vehicle, and
- the vehicle is one of
- an autonomous vehicle,
- a semi-autonomous vehicle, and
- a remotely controlled vehicle.
15. The aircraft carrier ground vehicle according to any one of claim 1, wherein
- the vehicle is a trailer equipped with a trailer coupler adapted to mechanically link to a coupling component of a powered tractor unit.
16. The aircraft carrier ground vehicle according to any one of claim 1, wherein
- the vehicle further designed to allow taxiing and take-off of the aircraft.
17. An aircraft carrier ground vehicle system including at least two aircraft carrier ground vehicles, each comprising:
- a chassis;
- a landing platform designed to accommodate an aircraft;
- a coupling mechanism, which connects the landing platform to the chassis to form a kinematic pair, in which the landing platform is rotatably mounted on the chassis;
- an actuation mechanism adapted to rotate the landing platform relative to the chassis, to allow an aircraft to land on the landing platform as the vehicle moves along a landing direction of the aircraft and compensate for an inclination angle of the aircraft on landing, in operation; and
- latching means adapted for gripping a structural part of the aircraft on landing, wherein the latching means includes a soft robotic suction cup latching mechanism.
18. The aircraft carrier ground vehicle system according to claim 17, wherein:
- each of the vehicles is a powered vehicle, which is remotely controlled; and
- the system further includes a central control unit that is configured to transmit control signals to each of the vehicles so as to
- rotate the landing platforms of the vehicles to compensate for an inclination angle of the aircraft on landing, and
- coordinate movements of the vehicles to allow the aircraft to land on the landing platforms of the vehicles as they move along the landing direction of the aircraft, in operation.
19. A method of landing an aircraft, the method comprising:
- providing an aircraft carrier ground vehicle comprising: a chassis; a landing platform designed to accommodate an aircraft; a coupling mechanism, which connects the landing platform to the chassis to form a kinematic pair, in which the landing platform is rotatably mounted on the chassis; an actuation mechanism adapted to rotate the landing platform relative to the chassis, to allow an aircraft to land on the landing platform as the vehicle moves along a landing direction of the aircraft and compensate for an inclination angle of the aircraft on landing, in operation; and latching means adapted for gripping a structural part of the aircraft on landing, wherein the latching means includes a soft robotic suction cup latching mechanism;

accelerating the aircraft carrier ground vehicle to adjust the speed of the vehicle to a landing speed of the aircraft, to bring the aircraft carrier ground vehicle at a landing position of the aircraft;

rotating the landing platform of the aircraft carrier ground vehicle to match an inclination angle of the aircraft; and landing the aircraft on the landing platform as the aircraft carrier ground vehicle moves along the landing direction of the aircraft and gripping a structural part of the aircraft thanks to said soft robotic suction cup latching mechanism.

20. An aircraft take-off method comprising:

providing an aircraft carrier ground vehicle comprising: a chassis; a landing platform designed to accommodate an aircraft; a coupling mechanism, which connects the landing platform to the chassis to form a kinematic pair, in which the landing platform is rotatably mounted on the chassis; an actuation mechanism adapted to rotate

the landing platform relative to the chassis, to allow an aircraft to land on the landing platform as the vehicle moves along a landing direction of the aircraft and compensate for an inclination angle of the aircraft on landing, in operation; and latching means adapted for gripping a structural part of the aircraft on landing, wherein the latching means includes a soft robotic suction cup latching mechanism;

loading and stowing an aircraft on the landing platform of the aircraft carrier ground vehicle;

accelerating the aircraft carrier ground vehicle to a take-off speed of the aircraft; and

releasing the aircraft for take-off with the landing platform rotated to a take-off configuration, after releasing said structural part of the aircraft from the soft robotic suction cup latching mechanism.

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